

When Yield Curves Invert Together: Predicting Currency Carry Losses

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August 2026

Job Market Paper

Abstract

National yield curves mix country-specific and shared information. I use their cross-country configuration to construct a predictive state: fresh 10-year/2-year inversions must be live in at least two non-U.S. G10 economies. From 1988 to 2026, annualized carry-trade spot returns are 12.9 percentage points lower when the lagged state is active, with inference based on fifteen episodes. Separate exactly-one and exactly-two state coefficients have opposite signs, and the configuration retains predictive content alongside the U.S. curve, the average G10 slope, and individual-country states. Losses fall on the high-interest-rate currencies bought by carry trades and grow with predetermined S&P 500 equity beta. The precise carry estimate depends on tenor and release design, while the exposure ordering is more stable. Cross-country yield-curve configurations therefore reveal a predictive state for exposure-ordered currency losses.

JEL: E43, F31, G12, G15.

Keywords: currency carry trade, yield-curve inversion, exchange-rate predictability, international bond markets.

1 Introduction

Currency portfolios diversify national shocks but remain exposed when several currencies move together. This distinction is central to the carry trade. Carry buys a standing set of high-interest-rate currencies and funds it with low-interest-rate currencies. The interest differential accrues gradually; one shared depreciation of the long leg can erase many months of income. This paper asks whether the cross-country configuration of bond markets reveals that loss state before the currency payoff.

One yield-curve inversion is ambiguous. It can reflect domestic weakness, country-specific disinflation, or a national term-premium movement. A cluster of fresh inversions is a different object: several bond markets cross the same economically salient boundary within overlapping windows. I construct a monthly state from 10-year/2-year slopes in nine non-dollar G10 economies. A fresh inversion observed in month $t - 1$ becomes live in month t and remains live until its slope has risen for two consecutive months. The state is active when at least two curves are live. The month- t state predicts the return in month $t + 1$.

The first result is a large conditional spot-return shift. The 1988–2026 sample contains 92 active months in fifteen episodes. Annualized carry spot returns average 2.85 percent outside the state and -10.00 percent inside it, a difference of -12.85 percentage points. The money-market-differential total-return proxy falls from 6.45 to -4.86 percent. The known interest differential remains positive, but it absorbs only part of the depreciation. The location shift reaches beyond the worst observations: the monthly spot median falls from 0.35 to -0.76 percent, and an outcome-conditioned bottom-decile deletion leaves a -7.24 -point spot difference.

The second result isolates the cross-country configuration. Separate indicator regressions give a positive coefficient for exactly one live inversion and a -15.72 -point annualized coefficient for exactly two; they do not provide a direct one-versus-two contrast. In joint regressions, the cluster retains predictive content alongside the U.S. inversion, the continuous average G10 slope and its change, and all nine country-level live states. The raw count and nearby release and tenor rules reveal meaningful sensitivity, not a universal cutoff: the 10Y–3M and three-month-release constructions attenuate the carry coefficient. The empirical object is therefore the nonlinear configuration of fresh crossings, with a precise historical estimate near the baseline rule, rather than “inversion” in general.

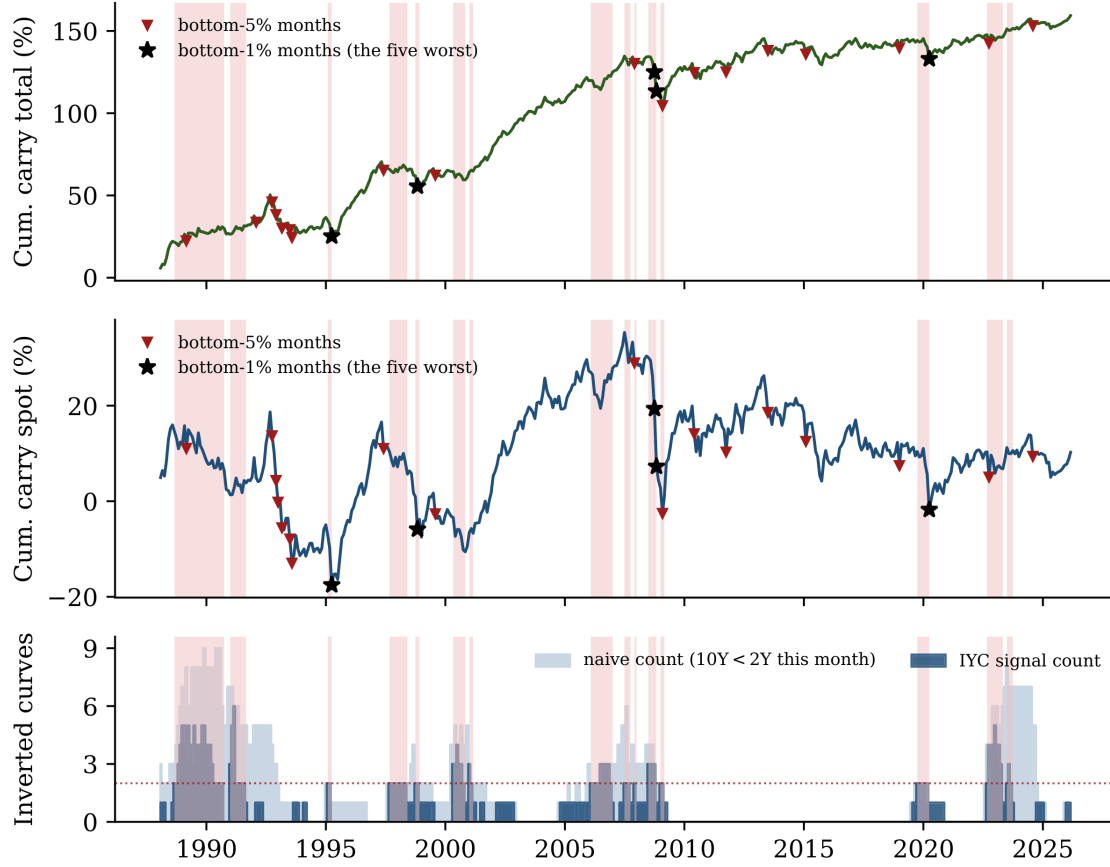


Figure 1: Carry returns and synchronized inversions. The light and dark lines count currently inverted and live curves; the dotted line marks the threshold of two. Shading is the month-end state, used with a one-month lag in return regressions.

The third result shows where the loss lands. High-interest-rate currencies depreciate relative to the low-rate currencies used to fund carry. Bilateral state coefficients also become more negative with currencies' predetermined equity-market betas. The cross-currency gradient explains 77 percent of the variation across nine coefficients; a portfolio sorted directly on expanding-window S&P 500 betas loses 7.53 percentage points per year in spot returns in the state. The exposure result is not uniform across every benchmark: an MSCI World beta sort is weak over the maximal full sample but agrees after 1999. This pattern makes the cross-sectional content sharper than a generic dollar or volatility state while keeping the beta measure itself contestable.

The paper's contribution is to change the unit of observation. National yield slopes forecast activity and combine expected short-rate paths with term premia (Harvey, 1988; Estrella and Hardouvelis, 1991; Estrella and Mishkin, 1998; Campbell and Shiller, 1991; Hamilton and Kim, 2002). International curves share important variation (Plosser and Rouwenhorst, 1994; Bernard and Gerlach, 1998; Diebold et al., 2008; Chinn and Kucko, 2015). Carry returns, in turn, load on common currency and stress factors (Fama, 1984; Brunnermeier et al., 2008; Lustig et al., 2011; Menkhoff et al., 2012; Farhi and Gabaix, 2016). The new empirical object is neither one country's slope nor a smooth global average. It is a cross-country threshold configuration, dated before the payoff and paired with a predetermined exposure gradient.

That configuration produces a disciplined economic interpretation. Write national slopes as a shared component plus country-specific information. Several fresh threshold crossings are more consistent with shared information than one crossing under restrictions on the idiosyncratic components. If currencies differ in their exposure to a subsequent adverse return component, the losses should be ordered across currencies. The exact-one/exact-two contrast, the reported curve-state horse race, and the beta gradient implement those predictions. They support a shared-information interpretation; they do not distinguish global growth, coordinated policy, inflation, term premia, or intermediary risk-bearing as the primitive shock.

Two design facts govern the strength of the claim. First, fifteen episodes supply the effective common-state variation. Second, the tenor, threshold, freshness rule, and release rule were assessed on the same historical sample. I therefore show the reported local state-definition neighborhood and add an independently executable public 10Y–3M audit that exhaustively reports a declared 64-rule family. The public family frequently yields negative coefficients, but no rule meets the 5-percent common-

calendar rotation-reference criterion. That adverse result places the ambition of the paper where the evidence is strongest: a novel cross-country configuration and exposure pattern, not an immaculate universal trading rule.

A public state that forecasts a negative total-return proxy also raises a pricing question. Predetermined carry compensates exposure only if it adjusts enough before the loss. A compact benchmark shows the restriction required for current expected losses to exceed both ordinary income and previously embedded compensation. It does not identify slow adjustment. The empirical contribution is prediction and exposure ordering; the friction remains an open mechanism.

Section 2 develops the framework, return accounting, state, and data. Section 3 defines estimation and inference. Section 4 establishes the cross-country configuration, and Section 5 shows where losses fall. Section 6 examines alternative explanations and the independent public audit. Section 7 states the public-information puzzle. Section 8 concludes. The online appendix gives the exact recursion, derivations, supplementary evidence and adverse robustness, and executable public-data checks.

2 Economic Framework and Measurement

2.1 Cross-country configurations as information

Let the 10-year/2-year slope in country i be

$$q_{i,t} = \mu_i + f_t + u_{i,t}, \quad (1)$$

where μ_i is a country level, f_t is information shared across curves, and $u_{i,t}$ collects country-specific information and measurement error. A lower f_t tends to flatten several curves. The inversion indicator is

$$I_{i,t} = \mathbf{1}\{q_{i,t} < 0\}. \quad (2)$$

One threshold crossing can arise from either component. Several fresh crossings are harder to attribute to one country's $u_{i,t}$. This signal-extraction logic requires restrictions on the cross-country dependence of the country-specific components: regional shocks, coordinated policy, or common measurement errors can also produce synchronization. Equations (1)–(2) therefore motivate empirical comparisons; they do not identify f_t or assign a structural posterior probability to two crossings.

The framework yields a nonlinear prediction. Exactly one inversion need not be a weaker adverse state than two. A single curve can identify a national rate cycle, whereas a cluster can change the composition of likely information. It also yields a spanning prediction: if the cross-country configuration matters, it should add information beyond one country's curve and a continuous average slope. The threshold and path rules are empirical measurements of that configuration, not primitives of the model.

The yield-slope component and the subsequent currency-return innovation are distinct. Write

$$\Delta s_{i,t+1} = -\lambda_i h_{t+1} + e_{i,t+1}, \quad (3)$$

where $h_{t+1} > 0$ is an adverse return component shared across currencies, λ_i is currency i 's exposure, and $e_{i,t+1}$ is idiosyncratic. The synchronized state is informative when $\mathbb{E}[h_{t+1} \mid S_t = 1]$ exceeds its inactive-state counterpart. Equation (3) then predicts larger depreciations for high- λ_i currencies. Predetermined equity beta is the empirical exposure proxy. The paper does not equate f_t with h_{t+1} ; the first is information in bond-market configurations, and the second organizes the currency payoff.

2.2 Currency returns and the shared-loss problem

Consider a U.S. investor holding currency i . Let $s_{i,t}$ be the log dollar price of one unit of foreign currency, so $\Delta s_{i,t+1} > 0$ is foreign-currency appreciation. Let $d_{i,t} \equiv (i_{i,t} - i_{US,t})/12$ be the one-month interest differential, formed from annualized money-market rates. The excess-return proxy is

$$rx_{i,t+1} = d_{i,t} + \Delta s_{i,t+1}. \quad (4)$$

The differential is known at t ; the spot change is realized from t to $t + 1$. Under covered interest parity this sum corresponds approximately to a forward return apart from basis and measurement differences (Du et al., 2018). It combines a simple annualized interest-rate differential divided by twelve with a log spot change, a first-order monthly approximation. Because the paper uses money-market differentials rather than executable forward quotes, the object is an asset-pricing return proxy, not a transaction-cost-adjusted strategy.

Each month, the carry portfolio buys the three highest-rate currencies and sells the three lowest-rate currencies among the nine non-dollar G10 currencies. If $\mathcal{H}_t$ and $\mathcal{F}_t$ denote the high-rate and funding sets, then

$$\begin{aligned} rx_{t+1}^{\text{carry}} &= \frac{1}{3} \sum_{i \in \mathcal{H}_t} rx_{i,t+1} - \frac{1}{3} \sum_{i \in \mathcal{F}_t} rx_{i,t+1} \\ &\equiv C_t + X_{t+1}. \end{aligned} \quad (5)$$

Here C_t is the long-minus-short interest differential and X_{t+1} is the corresponding spot return. The sample-average income leg is 3.9 percent per year and is positive in every month, with a minimum of 0.7 percent. Carry therefore loses only when the spot leg is more negative than the income leg is positive.

Diversification reduces $e_{i,t+1}$ but not the shared component when the long and funding legs have different average exposures. Here “shared” describes cross-currency comovement; “international” is the economic interpretation. The return equation is an organizing device rather than an estimated factor model.

2.3 The synchronized-inversion state

Let $q_{i,t} \equiv y_{i,t}^{10Y} - y_{i,t}^{2Y}$ denote country i 's end-of-month yield slope. An inversion can reflect expected easing, term-premium variation, or both (Campbell and Shiller, 1991; Hamilton and Kim, 2002; Bauer and Mertens, 2018).

Define $I_{i,t} = \mathbf{1}\{q_{i,t} < 0\}$. A crossing observed at $t - 1$, $I_{i,t-2} = 0$ and $I_{i,t-1} = 1$, makes the curve live at t . It remains live until $q_{i,t}$ has increased for two consecutive months. A released curve cannot re-enter while still inverted; it must first become nonnegative and then cross below zero again. Let $L_{i,t}$ be the resulting live status. The central state is

$$N_t = \sum_{i=1}^9 L_{i,t}, \quad S_t = \mathbf{1}\{N_t \geq 2\}. \quad (6)$$

Thus $L_{i,t}$ is path dependent, whereas $I_{i,t}$ depends only on the current slope. The state measured at the end of month t conditions the return during month $t + 1$.

The live rule has two economic roles. Delayed fresh entry excludes inversions already embedded in prices for many months and prevents the crossing-month return from entering the predictive state. Confirmed release allows the state to remain active as curves re-steepen, which can coincide with delivered easing and delayed currency adjustment. In the sample, a raw two-or-more current-inversion state is active in 165 months, compared with 92 for the live state, yet switches off before March 1995, October 2008, and March 2020—the three largest carry-loss months. This historical fit is informative and also creates specification-search risk. Online Appendix A states the transition rules and boundary cases exactly.

2.4 Data, information set, and predetermined exposure

The monthly sample runs from January 1988 through February 2026. The currency universe comprises the Australian dollar, Canadian dollar, Swiss franc, euro (Deutsche mark before 1999), British pound, Japanese yen, Norwegian krone, New Zealand dollar, and Swedish krona. The dollar is the numeraire and does not enter either the sort or the baseline state count. The U.S. curve is retained as a benchmark.

State incidence varies across monetary regimes. It is active in 33 of 84 months during 1988–94 but only 3 of 120 months during 2010–19, when many policy rates were near their lower bounds. The state is not a continuously available factor; it appears when international rate cycles create enough cross-country slope variation. Online Appendix

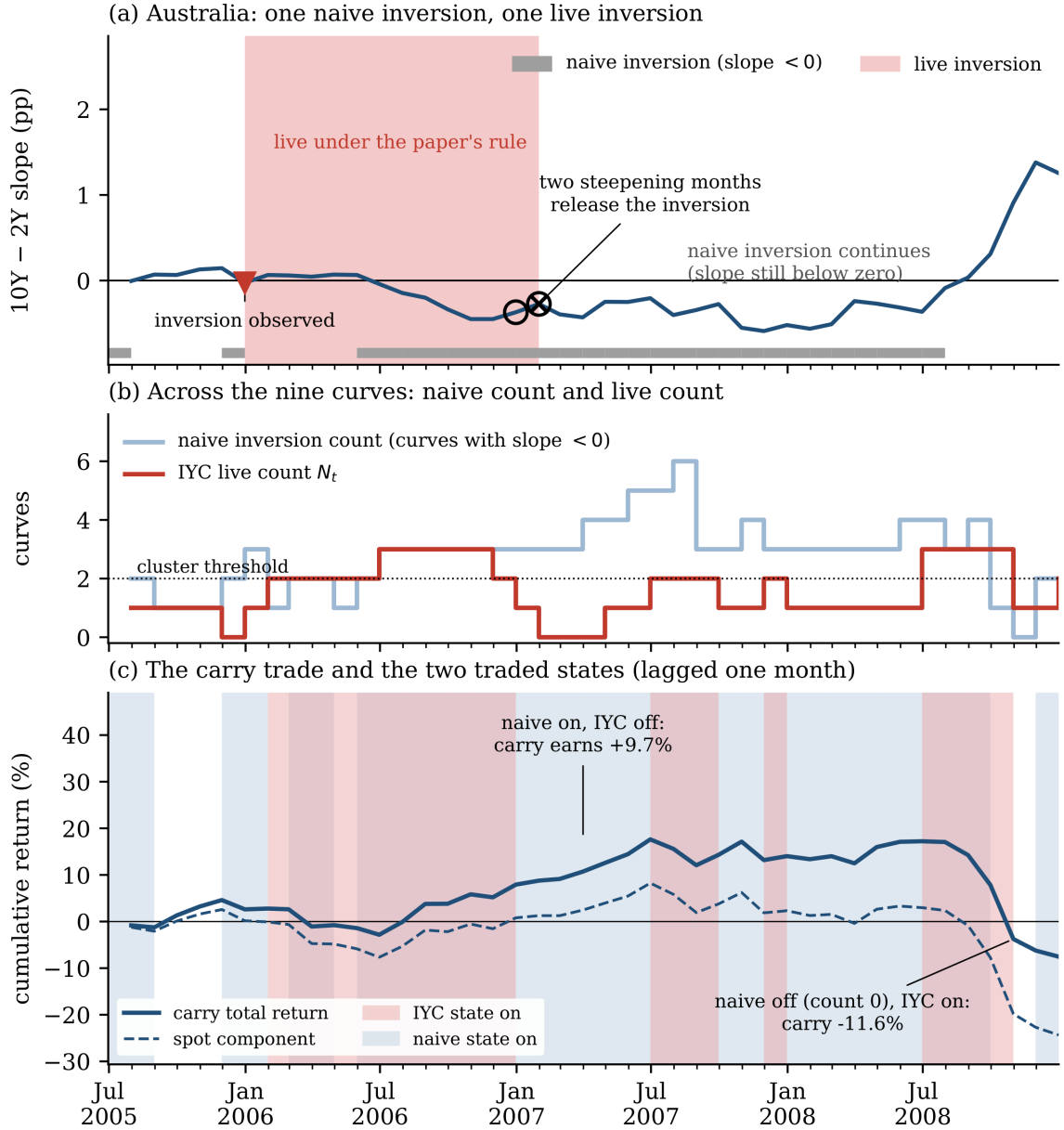


Figure 2: Live-state construction, 2005:07–2008:12. Panel (a) shows the Australian 10Y–2Y slope and its live status, which begins one month after the crossing is observed. Panel (b) compares the current-inversion and live counts. Panel (c) plots cumulative carry returns under the two lagged states. Delayed fresh entry removes the crossing-month return and stale inversions; confirmed release can retain the state after the current-inversion count declines.

A reports the full era counts.

The baseline state uses end-of-month 2-year and 10-year government yields. Portfolio weights use one-month money-market rates known at t . Return regressions use no information dated after t to predict month $t + 1$. This convention prevents mechanical look-ahead but does not eliminate asynchronous market closes, stale quotes, or later data revisions.

The exposure test estimates a currency’s equity beta on an expanding window:

$$rx_{i,\tau} = a_i + \hat{\beta}_{i,t} r_\tau^M + u_{i,\tau}, \quad \tau \leq t - 1. \quad (7)$$

The estimate $\hat{\beta}_{i,t}$ is available before the return from t to $t + 1$ and requires at least 36 prior monthly observations. The one-month buffer is part of the baseline construction: the estimate paired with S_t uses currency and equity returns only through $t - 1$. The beta is an empirical proxy for the latent exposure λ_i in equation (3); it need not equal that primitive. S&P 500 total returns form the market measure.

Table 1: Notation and timing

Symbol	Definition	Timing
$t; t + 1$	Information month-end; subsequent holding month	Information $\rightarrow$ return
$s_{i,t}; \Delta s_{i,t+1}$	Log dollar price of currency i ; its one-month change	Spot change realized in $t + 1$
$d_{i,t}$	Monthly foreign-minus-U.S. interest differential	Known at t
$rx_{i,t+1}$	Currency excess-return proxy	Realized in $t + 1$
$C_t; X_{t+1}$	Carry income leg; carry spot leg	Known at t ; realized in $t + 1$
$q_{i,t}$	Country i ’s 10Y–2Y yield slope	Observed at month-end t
$I_{i,t}; L_{i,t}$	Current inversion; path-dependent live inversion	Information at t
$N_t; S_t$	Live-curve count; synchronized-inversion state	Predict $t + 1$
$f_t; h_{t+1}$	Shared slope information; adverse currency-return component	Information at t ; payoff in $t + 1$
$\lambda_i; \hat{\beta}_{i,t}$	Latent return exposure; estimated equity-beta proxy	Beta uses data through $t - 1$
W_t	Vector of lagged benchmarks or controls	Known at t

Notes: Equations use the information-time convention $t \rightarrow t + 1$. Tables indexed by the return month equivalently report S_{t-1} .

3 Empirical Design

3.1 Predictive regression and inference

All return tests use one timing convention: information at t predicts the return at $t + 1$. The general specification is

$$Y_{t+1} = \alpha + bS_t + \Gamma'W_t + \varepsilon_{t+1}, \quad (8)$$

where W_t contains the relevant lagged benchmarks or controls. The headline conditional-mean difference sets W_t empty. With a binary state, b is the active-minus-inactive mean difference; monthly return coefficients are multiplied by twelve for reporting. In tables indexed by return month, equation (8) is equivalently written with S_{t-1} .

The state's 92 active months form fifteen contiguous episodes, fourteen longer than one month. Treating all active months as independent would overstate the information in the sample. I report Newey–West statistics with 12 lags and, for binary-state tests, a wild episode bootstrap that assigns Rademacher weights to contiguous episodes and imposes the null. The episode bootstrap is the headline inference because continuation months do not create independent synchronized shocks. The estimates can be economically large while their independent common-state variation remains fifteen episodes.

Distributional diagnostics report medians, lower quantiles, skewness, expected shortfall, and conditional variance. A separate diagnostic removes each outcome's full-sample bottom-decile observations from both states before re-estimating equation (8). Because that deletion conditions on realized returns, it is not a preferred estimator. It answers only whether a handful of extreme observations mechanically generate the mean gap.

3.2 Benchmarks and controls

The benchmark hierarchy moves from nearby measurement alternatives to broader risk states. The raw current-inversion state asks whether fresh entry and confirmed release add information. The U.S. inversion asks whether the conventional single-country proxy spans the international state. The average G10 slope and its change ask whether the threshold is merely a nonlinear transformation of a smooth global slope. Nine country-specific live states ask whether synchronized crossings add information beyond

their individual components.

Lagged NFCI and VIX controls address predictive spanning: do variables already known at t absorb the state’s forecast? Contemporaneous dollar and currency-volatility factors answer a different question: do those realized factors account for the loss after it occurs? Neither design identifies the shock represented by S_t . Joint specifications are reported without sequential orthogonalization because the ordering of closely related yield-curve transformations would otherwise determine which variable receives their shared variation.

3.3 Exposure ordering

The bilateral state coefficients come from

$$\Delta s_{i,t+1} = a_i + b_i S_t + e_{i,t+1}. \quad (9)$$

The nine estimates $\hat{b}_i$ are compared with currencies’ average predetermined equity betas. The slope, fitted intercept at zero beta, and uncertainty describe the exposure gradient. A beta-sorted portfolio and a panel interaction $S_t \times \hat{\beta}_{i,t}$ provide complementary tests: the portfolio is economically transparent; the panel uses monthly variation but still has only fifteen common-state episodes.

This design separates three claims. Prediction asks whether S_t forecasts returns. Exposure accounting asks whether the realized losses are ordered by a predetermined market characteristic. Mechanism identification would require independent variation that distinguishes global growth, policy, term-premium, inflation, and intermediary channels. The first two are directly estimated; the third is not.

3.4 Specification-family discipline

Two different specification exercises appear in the paper and are never pooled. The first is the state-definition neighborhood reported for the baseline 10Y–2Y analysis. It varies breadth, release confirmation, tenor, and current-count repairs. These rows reveal where the historical result strengthens or attenuates, but they are selected displayed specifications rather than the complete set of rules examined. Their p -values are per specification; the joint resamples required for family adjustment are unavailable.

The second exercise is an independently executable public proxy. Its code declares

and exhaustively reports 64 combinations of raw or live 10Y–3M states, breadth thresholds, release lengths, one- or two-month return lags, and target leg sizes of two or three, with all currencies tied at a cutoff included and equally weighted. It reports every coefficient, an inclusive circular-shift reference, and a common-calendar maximum-absolute-standardized-coefficient reference across the family. Under the maintained null that simultaneous cyclic shifts of the states relative to returns are exchangeable, this construction controls the declared family; without that assumption it is a finite rotation-reference diagnostic. The declaration was not preregistered before the data were examined. The public inputs and outcome differ from the baseline analysis, so this exercise measures sign prevalence and selection sensitivity rather than reproducing the headline estimate.

The evidentiary hierarchy follows from those designs. The baseline state supplies the economic result and cross-sectional ordering. Its displayed neighborhood shows known sensitivity. The public family shows what happens when a nearby declared rule family is exhaustively reported and jointly evaluated. A future observation under a dated, unchanged 10Y–2Y rule remains the clean prospective test.

3.5 Claims and rejection patterns

Four findings would run against the paper’s interpretation: the distribution shift vanishing outside a few tail observations; the U.S. or average-slope measures spanning the configuration; no cross-sectional relation between losses and predetermined exposure; or a public nearby-rule family with no sign resemblance. The reported tests run against the first two patterns. The S&P-based exposure tests run against the third, although the nine-point fit lacks a displayed confidence region, the panel interaction is imprecise, and the full-sample MSCI sort fails. The public evidence is weaker: negative coefficients are frequent, but no rule meets the 5-percent common-calendar rotation-reference criterion. That result preserves the cross-country-configuration question while ruling out a claim that the exact carry coefficient is invariant to reasonable measurement choices.

4 The Cross-Country Configuration

4.1 Conditional carry returns

Carry spot returns are 12.85 percentage points per year lower when the lagged synchronized-inversion state is active. They average 2.85 percent in 366 inactive months and -10.00 percent in 92 active months. The money-market-differential total-return proxy falls from 6.45 to -4.86 percent, a difference of -11.31 points. Episode-bootstrap p -values are 0.001 and 0.003.

Table 2: The carry trade in and out of the IYC signal

Portfolio component	Annualized mean		State regression	
	No signal	Signal	Difference	Episode wild p
Carry spot	2.85	-10.00	-12.85	0.001
Carry total	6.45	-4.86	-11.31	0.003
High-beta spot	1.26	-6.27	-7.53	0.010
High-beta total	3.18	-3.64	-6.82	0.013

Notes: Returns are annualized percentages. The no-signal state contains 366 months and the signal state 92 months over 1988:01–2026:02. The difference is the coefficient on the one-month-lagged IYC-state indicator. Wild-bootstrap p -values cluster by the state’s fifteen contiguous episodes ($B = 9,999$).

The accounting locates the result. The carry income leg averages 3.91 percent per year and is positive in every month, with a minimum of 0.7 percent. It does not fall enough to create the conditional loss; the spot depreciation overwhelms it. The object is therefore a predictable spot-risk state and, under the money-market accounting, a negative conditional total-return proxy. Executable forward quotes, basis, spreads, and financing costs are not observed over the full sample.

The distribution shifts with the mean. The monthly carry-spot median changes from 0.35 percent outside the state to -0.76 percent inside it. The 25th percentile moves from -0.93 to -1.98 percent and the 10th percentile from -2.35 to -3.94 percent. Expected shortfall worsens from -4.80 to -8.14 percent. The conditional variance ratio is 1.74, with permutation $p = 0.14$: the evidence is stronger for a leftward shift than for a general variance increase.

Deleting each series’ full-sample bottom-decile months leaves carry spot and total differences of -7.24 and -5.50 points, with episode-bootstrap $p = 0.006$ and 0.024. This deletion uses realized outcomes and is not a preferred estimator. Its narrower

purpose is clear: a small set of worst observations does not arithmetically generate the entire conditional mean.

4.2 Exactly-one and exactly-two indicator regressions

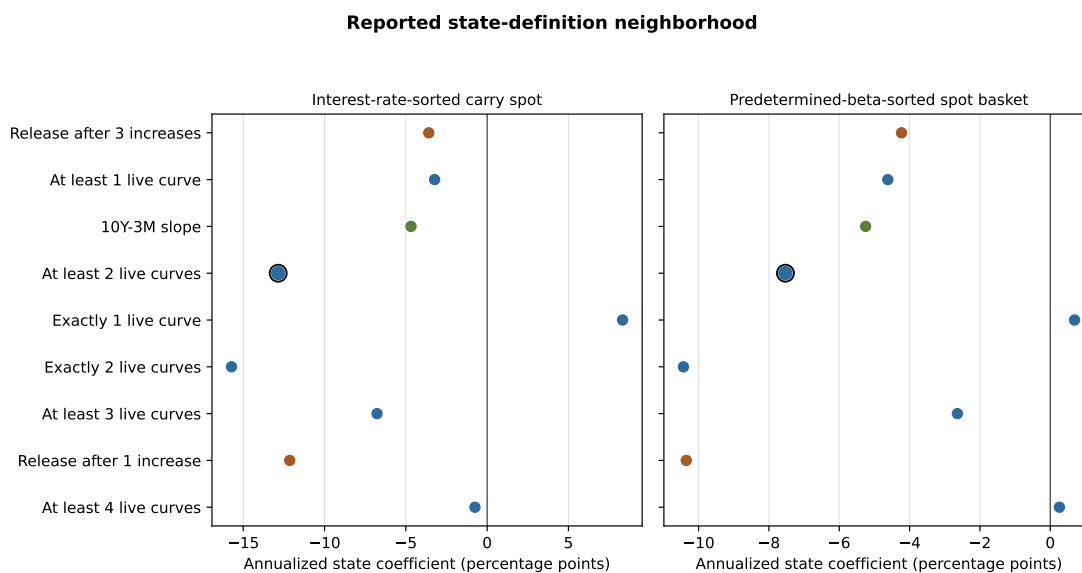
The most revealing comparison is inside the count. In 88 months with exactly one live inversion, carry spot and total coefficients are +8.33 and +8.78 percentage points per year, with $p = 0.005$ and 0.003. In 48 months with exactly two, the coefficients reverse to -15.72 and -15.13 , with $p = 0.001$ and 0.002. The S&P-beta portfolio shows the same qualitative break: +0.69 after exactly one and -10.43 after exactly two.

The opposite coefficient signs are the economic content of synchronization. A lone fresh inversion can arise from one country’s rate cycle, while a diversified carry portfolio continues to collect the cross-country interest differential. At the second curve, the measured configuration changes. Because each coefficient compares its indicated months with a different complement, these regressions do not estimate a direct exactly-one-versus-exactly-two contrast. They nevertheless show why an “any inversion” label can combine return associations with opposite signs. The estimates do not make two a universal structural threshold.

Counts above two do not produce a monotone dose response. At least three live curves gives a -6.77 -point carry-spot coefficient, and at least four gives -0.75 ; both are imprecise and use fewer months. These states also occur later within episodes. The nested at-least-two rule retains the configuration after additional curves enter; the exact-two row is the sharper historical cell, not a stable rule that switches off when a third curve inverts.

4.3 The reported state-definition neighborhood

Figure 3 displays the nine unique breadth, release, and tenor rules reported in the analysis. Rules are ordered by active-month count, independently of their return coefficient. Eight of nine carry-spot estimates and seven of nine S&P-beta spot estimates are negative. The baseline is the second most negative carry rule and the third most negative beta rule. It is not the single best displayed specification.



Rules are ordered by active-month count, not by estimated return. The black ring marks the baseline.
The displayed specifications do not exhaust the search and do not use multiplicity-adjusted inference.

Figure 3: Reported state-definition neighborhood. Points show the displayed breadth, release, and tenor specifications and are ordered by active months. The black ring marks the 10Y-2Y, at-least-two, two-month-release baseline. The display is not the complete set of rules examined and does not provide multiplicity-adjusted inference.

Table 3: Reported state-definition neighborhood

Rule	Active months	Carry spot b [p]	Carry total b [p]	High-beta spot b [p]	High-beta total b [p]
<i>Panel A: number of live inversions</i>					
≥ 1 (any live inversion)	180	-3.23 [0.230]	-1.90 [0.513]	-4.62 [0.008]	-4.16 [0.014]
$= 1$ (exactly one)	88	+8.33 [0.005]	+8.78 [0.003]	+0.69 [0.818]	+0.66 [0.831]
≥ 2 (baseline)	92	-12.85 [0.001]	-11.31 [0.003]	-7.53 [0.010]	-6.82 [0.013]
$= 2$ (exactly two)	48	-15.72 [0.001]	-15.13 [0.002]	-10.43 [0.012]	-10.31 [0.011]
≥ 3	44	-6.77 [0.405]	-4.57 [0.694]	-2.64 [0.478]	-1.47 [0.711]
≥ 4	22	-0.75 [0.879]	+2.10 [0.672]	+0.26 [0.967]	+1.87 [0.736]
<i>Panel B: release confirmation</i>					
One steepening month	39	-12.14 [0.059]	-10.49 [0.112]	-10.35 [0.035]	-9.39 [0.051]
Two consecutive months (baseline)	92	-12.85 [0.001]	-11.31 [0.003]	-7.53 [0.010]	-6.82 [0.013]
Three consecutive months	191	-3.59 [0.161]	-2.11 [0.491]	-4.23 [0.001]	-3.58 [0.017]
<i>Panel C: tenor and downcurve filter</i>					
10Y-2Y baseline	92	-12.85 [0.001]	-11.31 [0.003]	-7.53 [0.010]	-6.82 [0.013]
10Y-3M	107	-4.68 [0.146]	-4.02 [0.260]	-5.25 [0.011]	-5.00 [0.034]
10Y-2Y, excluding downcurve	57	-18.07 [0.001]	-17.61 [0.002]	-12.89 [< 0.001]	-12.59 [0.001]
10Y-3M, excluding downcurve	74	-5.73 [0.177]	-6.10 [0.201]	-7.56 [0.003]	-7.94 [0.004]
<i>Panel D: current-count repairs</i>					
Current count ≥ 2	—	-3.18 [0.047]	-1.06 [0.670]	-4.73 [0.005]	-4.06 [0.012]
Current count, excluding downcurve	—	-3.31 [0.238]	-1.73 [0.581]	-6.43 [0.003]	-6.13 [0.006]
Current count, reciprocal-age weighted	—	-10.43 [0.040]	-9.75 [0.057]	-14.87 [0.001]	-14.46 [0.002]
Live state (baseline)	—	-12.85 [0.001]	-11.31 [0.003]	-7.53 [0.010]	-6.82 [0.013]
Live state, excluding downcurve	—	-18.07 [0.001]	-17.61 [0.002]	-12.89 [< 0.001]	-12.59 [0.001]

Notes: Annualized active-minus-inactive coefficients with the wild-bootstrap p -value on each rule's episodes in brackets; 1988:01–2026:02. These are the displayed breadth, release, tenor, and current-count variants, not a complete record of researcher discretion; search-adjusted or family-wise p -values are unavailable. The two tenor states overlap in 58 months. The 10Y–3M downcurve stay-set has carry total +1.59 with $p = 0.371$. Under NFCI/VIX controls, the current count has carry spot -1.91 [0.576] and total -0.63 [0.847]; the reciprocal-age rule has carry spot -7.55 [0.084] and high-beta spot -13.82 [0.008]; and the downcurve-filtered current count has carry spot -3.99 [0.269] and high-beta spot -7.04 [0.035]. Active-month counts are not displayed for Panel D and are left blank.

The carry estimate has two load-bearing measurement margins. Releasing after one increase leaves a large -12.14 -point spot estimate but less episode-level precision ($p = 0.059$); releasing after three increases expands the state to 191 months and attenuates the estimate to -3.59 ($p = 0.161$). Replacing the 2-year yield with a 3-month rate produces -4.68 ($p = 0.146$). The S&P-beta spot estimates remain negative and statistically detectable under both perturbations. The state is therefore more successful at locating an exposure-ordered currency condition than at delivering an invariant carry magnitude.

The raw current-inversion count supplies a second benchmark. Its carry spot coefficient is -3.18 ($p = 0.047$), compared with -12.85 for the live state. Weighting current inversions by reciprocal age revives the coefficient to -10.43 ($p = 0.040$) and produces a -14.87 S&P-beta estimate ($p = 0.001$). Freshness contains information. The live release rule adds a different timing margin by remaining active as curves begin to re-steepen; that margin also retains several historically large loss months. This is

exactly why the public rule-family audit in Section 6 matters.

4.4 Synchronization versus nearby curve states

The portfolio horse race tests whether the configuration adds information beyond the displayed alternatives. The first joint specification includes the continuous average G10 10Y–2Y slope, its one-month change, and the U.S. inversion. The synchronized-state coefficients remain -13.45 for carry spot and -12.44 for total returns, with episode-bootstrap $p = 0.003$ and 0.012 . The average-slope variables are imprecise in the carry columns, and the U.S. inversion enters positively and imprecisely.

Adding all nine country-level live states makes the synchronized coefficient more negative: -26.79 for carry spot and -27.52 for total returns. The nine own states are jointly informative as well. The larger cluster coefficient is not a causal purification. The related regressors divide variation from the same curves, and the own states absorb many benign single-country months. The appropriate conclusion is that the cross-country configuration contains information not spanned by the reported continuous, U.S., or own-state transformations.

Table 4: The synchronized configuration versus reported portfolio-level rivals

Regressor	Specification (2)			Specification (3)		
	b	t_{NW}	Wild p	b	t_{NW}	Wild p
<i>Panel A: carry spot</i>						
Cluster	−13.45	−3.15	0.003	−26.79	−4.88	< 0.001
Global slope	+0.21	+0.08	—	+4.68	+1.28	—
Δ global slope (1m)	−14.97	−1.15	—	−16.10	−1.23	—
U.S. inversion	+2.95	+0.74	0.465	+2.23	+0.51	0.605
Nine own signals (joint p)	—	—	—	—	—	0.005
<i>Panel B: carry total</i>						
Cluster	−12.44	−2.86	0.012	−27.52	−4.92	< 0.001
Global slope	−0.64	−0.22	—	+4.30	+1.17	—
Δ global slope (1m)	−15.14	−1.17	—	−16.22	−1.24	—
U.S. inversion	+2.42	+0.60	0.581	+1.63	+0.37	0.696
Nine own signals (joint p)	—	—	—	—	—	< 0.001
<i>Panel C: high-beta spot</i>						
Cluster	−6.16	−1.72	0.102	−11.99	−2.02	0.040
Global slope	+2.40	+0.90	—	+5.45	+1.58	—
Δ global slope (1m)	+11.87	+0.92	—	+10.85	+0.87	—
U.S. inversion	+1.81	+0.72	0.450	+1.27	+0.38	0.667
Nine own signals (joint p)	—	—	—	—	—	0.009
<i>Panel D: high-beta total</i>						
Cluster	−5.66	−1.56	0.143	−12.37	−2.07	0.036
Global slope	+2.40	+0.88	—	+5.67	+1.62	—
Δ global slope (1m)	+11.41	+0.89	—	+10.32	+0.83	—
U.S. inversion	+2.57	+1.00	0.308	+2.00	+0.57	0.543
Nine own signals (joint p)	—	—	—	—	—	0.004

Notes: Sample: 1988:01–2026:02. Specification (2) includes the cluster, the average G10 10Y–2Y slope, its one-month change, and the lagged U.S. inversion. Specification (3) adds all nine country-level live states. State coefficients are annualized percentage points; slope coefficients are annualized percentage points per percentage-point slope. Continuous regressors report Newey–West statistics only. Binary-state p -values use a wild bootstrap on the relevant state’s episodes, with other regressors partialled under the null. The nine own-state coefficients are not displayed individually; their joint p -value is shown. Cluster-alone coefficients appear in Table 2. U.S. inversion alone: carry spot −1.90 [0.484], carry total −1.34 [0.637], high-beta spot −3.01 [0.205], and high-beta total −2.03 [0.345].

The U.S. inversion alone reinforces the distinction. It is active in 57 lagged months and has carry spot and total coefficients of −1.90 ($p = 0.484$) and −1.34 ($p = 0.637$). Adding the U.S. curve to the cluster count, however, leaves the baseline result nearly unchanged. The U.S. curve is neither a substitute for the international configuration nor a reason to treat the United States as economically irrelevant.

5 Where the Losses Fall

5.1 Interest rates and predetermined equity exposure

The carry result contains a cross-sectional ordering before any beta is estimated. The portfolio loses because the high-interest-rate currencies it buys depreciate relative to the low-interest-rate currencies used for funding. The bilateral regression

$$\Delta s_{i,t+1} = a_i + b_i S_t + e_{i,t+1}$$

asks whether a second predetermined characteristic orders the nine currency coefficients.

Australia and New Zealand have average S&P equity betas of 0.20 and 0.13 and annualized state spot coefficients of -13.21 and -13.28 percentage points. Japan and Switzerland have betas near zero and coefficients of $+2.35$ and -0.89 . Regressing the nine state coefficients on average predetermined beta produces a slope near -60 percentage points per beta unit, an intercept of -0.75 , and $R^2 = 0.77$.

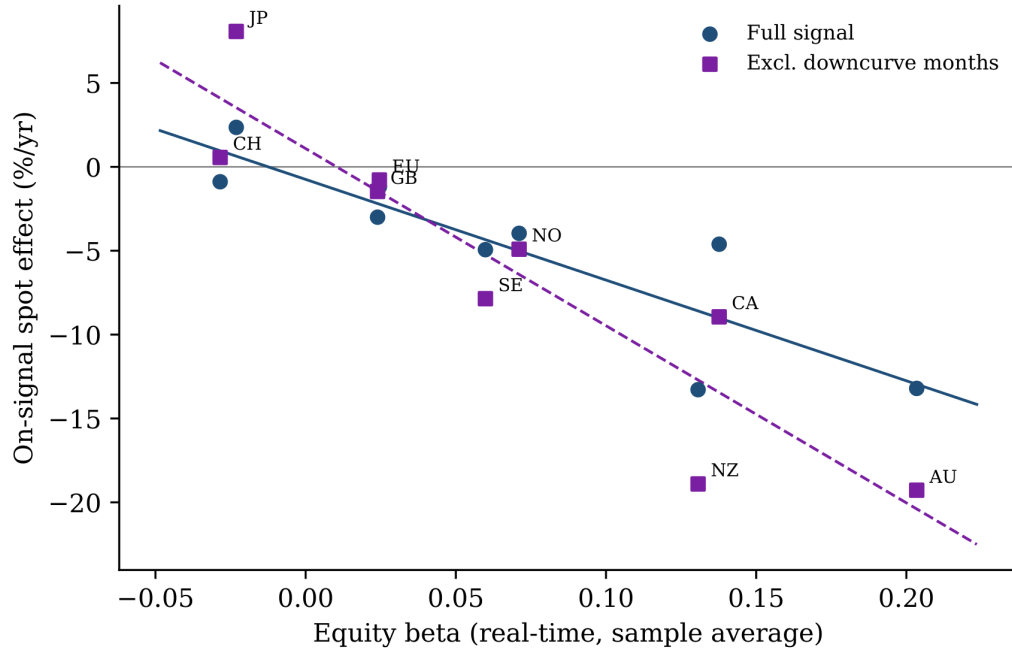


Figure 4: Currency state coefficients and predetermined equity beta. Each point is one bilateral spot-return coefficient. The solid line is the baseline fit (slope -60 , intercept -0.75 , $R^2 = 0.77$); the dashed fit excludes the exploratory downcurve months.

The nine-point gradient is economically sharp and statistically limited. It has no

displayed confidence region for the fitted intercept, and Australia and New Zealand exert substantial leverage. It does not establish a structural beta or prove that exposure vanishes at zero. It documents a descriptive restriction that a uniform dollar shift does not match: in the S&P-based ranking, currencies with greater predetermined market exposure have larger state-conditioned losses.

Table 5: How active-state losses align with exposure

Object	Estimate	Episode wild p	Interpretation
Rate-beta rank correlation	0.72	—	Descriptive, nine currencies
Cross-currency beta slope	−60	—	Percentage points per beta unit
Cross-currency fitted intercept	−0.75	—	Percentage points at zero beta
Cross-currency fit	$R^2 = 0.77$	—	Descriptive, nine currencies
Beta-sorted spot portfolio	−7.53	0.010	Annualized state difference
Panel state $\times$ beta	−27	0.210	Annualized interaction
Conditional Fama slope, active state	−1.33	0.340	p -value is for the state interaction

Notes: The cross-currency slope, intercept, and R^2 come from regressing the nine bilateral state coefficients on currencies' sample-average predetermined equity betas. The beta-sorted portfolio uses expanding-window betas. The panel interaction uses monthly predetermined betas. The Fama row is supporting conditional-moment evidence; its episode-level uncertainty is wide.

The beta-sorted portfolio supplies a portfolio-level implementation of that restriction. It buys the three currencies with the highest expanding-window betas and sells the three lowest, using estimates available before the return. Its spot return is 7.53 percentage points per year lower in the state, with episode-bootstrap $p = 0.010$. The monthly panel interaction between the state and predetermined beta is also negative, about −27 points per beta unit, but its episode-bootstrap p -value is 0.210. The portfolio and bilateral gradient carry the exposure evidence; the panel interaction shows that its precision should not be inferred from the number of currency-month observations.

5.2 The cluster and each currency's own curve

Country-level regressions compare four objects: a currency's own live state alone, the synchronized cluster alone, both together, and a full specification adding own-slope variables and the average G10 slope. The cluster in these regressions still includes the

predicted country’s curve. This is an own-curve horse race, not a leave-one-country-out design.

The Australian own state is the only own-state coefficient detected on its own: -8.76 points with $p = 0.044$. Conditional on the cluster, every own-state coefficient is imprecise. Cluster-alone estimates are -13.21 for Australia ($p = 0.015$) and -13.28 for New Zealand ($p = 0.011$). In the full specification their point estimates remain negative, at -12.92 and -8.56 , but episode-bootstrap precision weakens. Across the pooled bilateral panel, the cluster coefficient is -4.74 with Newey–West $t = -2.34$, while the pooled own-state coefficient is -0.01 with $t = -0.01$.

Own-curve controls are jointly detected for Japan and not for the other eight currencies. This is an economically useful exception: the bundle of Japanese own-state, relative-slope, and relative-level controls is jointly detected even though the yen’s cluster coefficient is positive. The joint test does not isolate which own-curve component supplies the information. More generally, the own-curve specifications do not reproduce the adverse long-minus-short portfolio pattern. A stronger claim—that other countries’ curves predict a currency after its own curve is excluded—requires the unavailable country-month 10Y–2Y state matrix.

5.3 Persistent portfolio cores with some rotation

The rate and beta sorts hold persistent currency blocs. New Zealand occupies the carry long leg in 85 percent of all months and 91 percent of active-state months; Australia does so in 71 and 73 percent. Japan is in the funding leg in 83 percent of all months and 97 percent of active months; Switzerland’s corresponding shares are 96 and 100 percent. For the beta sort, Canada is always in the long leg, Australia is long in 97 percent of all months and 92 percent of state months, and New Zealand in 77 and 76 percent.

The state-conditioned portfolios retain persistent cores, so the result is not created solely by resorting when the signal turns on. There is meaningful rotation around those cores: Norway’s carry-long share falls from 62 percent overall to 34 percent in state months, while the pound’s rises from 43 to 70 percent. Interest rates matter twice: they deliver known carry income and identify a persistent long-leg core that bears the spot loss. The sample rank correlation between average rate and beta rankings is 0.72. Country size, commodity exposure, trade position, and intermediary demand can all

contribute to that alignment (Hassan, 2013; Ready et al., 2017; Della Corte et al., 2016; Richmond, 2019; Hassan and Mano, 2019; Gabaix and Maggiori, 2015). The analysis does not select a unique primitive; it shows that two predetermined rankings point toward the same loss-bearing currencies.

Table 6: Persistent portfolio cores and state-dependent rotation

Currency	Carry portfolio		Equity-beta portfolio	
	Long	Funding	Long	Funding
New Zealand	85 / 91	2 / 1	77 / 76	0 / 0
Australia	71 / 73	4 / 0	97 / 92	0 / 0
Norway	62 / 34	3 / 5	26 / 32	0 / 0
United Kingdom	43 / 70	4 / 0	0 / 0	52 / 47
Canada	20 / 21	23 / 14	100 / 100	0 / 0
Japan	0 / 0	83 / 97	0 / 0	73 / 65
Switzerland	0 / 0	96 / 100	0 / 0	95 / 85

Notes: Entries are the percentage shares of months in which a currency appears in the indicated three-currency leg, reported as full sample / active state. The full sample contains 458 portfolio months and the active state contains 92. The table selects seven currencies from the complete nine-currency composition table in the Online Appendix.

5.4 How stable is the beta ordering?

Across the reported state-definition neighborhood, the S&P-beta portfolio is more stable than carry. Its spot coefficient is -10.35 under one-month release, -4.23 under three-month release, and -5.25 under the 10Y–3M construction; all three are statistically detected at the reported levels. Removing fitted exposure to the U.S. ACM slope term premium leaves a -6.17 -point beta coefficient with $p = 0.004$, even as the carry coefficient attenuates.

The exposure measure itself has an important failure. Rebuilding the portfolio with expanding MSCI World betas produces a full-sample spot coefficient of -2.93 with $p = 0.319$, compared with -7.93 and $p = 0.005$ for S&P betas on the same maximal sample. After 1999, both definitions are negative and detected. Early Japanese equity weight can explain why an external U.S. index and a capitalization-weighted world index rank the yen differently, but it does not erase the full-sample benchmark dependence.

Table 7: Equity-beta benchmark sensitivity

Sample	Beta benchmark	N	Spot b	Wild p	Total b	Wild p
<i>Panel A: synchronized-inversion state</i>						
Maximal all-nine-beta	S&P 500	443	-7.93	0.005	-7.19	0.009
Maximal all-nine-beta	MSCI World	443	-2.93	0.319	-4.00	0.162
Post-1999	S&P 500	320	-11.10	0.003	-10.73	0.004
Post-1999	MSCI World	320	-7.67	0.030	-7.85	0.019
<i>Panel B: synchronized-inversion state excluding downcurve months</i>						
Maximal all-nine-beta	S&P 500	443	-12.31	0.001	-11.97	0.001
Maximal all-nine-beta	MSCI World	443	-2.30	0.544	-2.65	0.446
Post-1999	S&P 500	320	-13.35	0.001	-13.09	0.001
Post-1999	MSCI World	320	-8.29	0.014	-8.67	0.008

Notes: Annualized percentage-point state coefficients for high-minus-low portfolios sorted each month on expanding currency betas estimated against the named equity index; sorts are lagged one month. Panel A uses each definition's maximal all-nine-beta sample and therefore excludes fifteen early months in which the baseline portfolio uses seven or eight available betas. The full-sample MSCI-based result is not statistically detectable. Japan's declining weight in MSCI World helps explain the post-1999 agreement but does not remove the full-sample benchmark sensitivity.

The state forecasts losses in the high-rate leg, and a predetermined S&P-beta rank orders them across currencies and in a separate portfolio. This benchmark-sensitive gradient turns a time-series gap into a testable statement about who bears it; it is not a structural mechanism.

6 Alternative Explanations and Independent Stress Tests

6.1 The United States, financial conditions, and volatility

The U.S. curve does not substitute for the cross-country configuration. Its inversion alone has a -1.90 -point carry-spot coefficient with $p = 0.484$. Adding it to the configuration count is a different experiment. The resulting ten-curve state contains all 92 baseline months plus nine additional months, and the carry-spot coefficient remains -11.72 ($p = 0.001$). Including the dollar in the portfolio also leaves the estimate near -13 points. The exclusion of the United States is therefore not load-bearing for the reported association, even though the U.S. curve contributes little on its own.

Table 8: Including the U.S. curve and the dollar

Portfolio	Signal construction	Return leg	b	t_{run}	Wild p
G10 carry	Baseline (nine curves)	Spot	-12.85	-3.03	0.001
G10 carry	Baseline (nine curves)	Total	-11.31	-2.40	0.003
G10 carry	Add U.S. curve (ten curves)	Spot	-11.72	-2.92	0.001
G10 carry	Add U.S. curve (ten curves)	Total	-10.20	-2.34	0.006
Dollar-inclusive	Baseline (nine curves)	Spot	-13.11	-2.69	< 0.001
Dollar-inclusive	Baseline (nine curves)	Total	-11.54	-2.15	0.009
Dollar-inclusive	Add U.S. curve (ten curves)	Spot	-12.10	-2.68	0.001
Dollar-inclusive	Add U.S. curve (ten curves)	Total	-10.57	-2.14	0.013

Notes: Coefficients are annualized percentage-point state effects from bare state regressions; t_{run} clusters by signal run and p is a wild bootstrap on signal runs ($B = 9,999$). The baseline state has 92 months. Adding the U.S. curve produces 101 state months and contains all 92 baseline months. The dollar sits in the dollar-inclusive portfolio’s funding leg in 31 percent of months. The result does not depend on excluding the U.S. from either margin; this does not make the U.S. curve structurally irrelevant.

Lagged financial conditions do not absorb the baseline state. With the NFCI level and change, the carry-spot coefficient is -12.84 points ($p < 0.001$). Adding lagged VIX level and change gives -13.18 ($p = 0.004$). Contemporaneous dollar and realized FX-volatility controls leave carry spot at -11.5 ($p = 0.002$) and the S&P-beta portfolio at -7.4 ($p = 0.009$). These designs answer different questions. Lagged variables test predictive spanning; contemporaneous variables account for measured features of the realized loss.

The favorable realized-volatility result is not universal. Substituting G10 implied-

volatility innovations leaves the carry estimate statistically detected but produces high-beta coefficients near -7.5 with p -values around 0.17 – 0.18 . The raw current-inversion count is also absorbed by the full NFCI/VIX control set. These failures reinforce the comparative result: the live configuration adds information beyond the displayed lagged conditions, but neither every volatility proxy nor every curve-state construction preserves the same cross-sectional precision.

6.2 Expected rates and a common term-premium proxy

An inversion combines expected short-rate news and term-premium variation. In a predictive-spanning exercise, adding the lagged U.S. ACM slope-premium level, or its level and one-month change, leaves the carry-spot coefficient near -12.9 percent ($p = 0.001$) and the S&P-beta spot coefficient near -7.6 percent ($p = 0.011$). A separate decomposition removes each country’s fitted exposure to the U.S. ACM 10Y-minus-2Y slope term premium, then rebuilds the state. Eighty-three percent of national inversion observations remain. The stripped state is broader—125 months in eighteen runs—and overlaps the baseline in 83 months.

The carry result attenuates: the spot coefficient falls from -12.85 to -5.42 and has $p = 0.107$. The S&P-beta coefficient remains -6.17 with $p = 0.004$. The rebuilt state still contains all five worst carry months but expands from 92 to 125 active months. Its smaller carry coefficient therefore reflects a changed classifier—and possible dilution from the additional months—as well as removal of fitted exposure to the U.S. premium. The exercise does not identify what share of the baseline association is attributable to a common term-premium component, and one U.S. proxy cannot separate expected rates and term premia country by country.

Table 9: State rebuilt after removing fitted exposure to the U.S. ACM slope term premium

State	Carry spot b [p]	Carry total b [p]	High-beta spot b [p]	High-beta total b [p]
Baseline state	−12.85 [0.001]	−11.31 [0.003]	−7.53 [0.010]	−6.82 [0.013]
Stripped state	−5.42 [0.107]	−4.07 [0.251]	−6.17 [0.004]	−5.66 [0.008]
Non-downcurve months	−7.68 [0.094]	−7.03 [0.142]	−10.27 [0.000]	−10.28 [0.000]
Downcurve months	+1.56 [0.621]	+3.81 [0.059]	+5.00 [0.026]	+6.36 [0.012]
Downcurve wedge within stripped state	+7.64 [0.149]	+9.18 [0.094]	+12.89 [0.007]	+14.16 [0.004]

Notes: Sample: 1988:01–2026:02. Each national 10Y–2Y slope is regressed on the U.S. ACM 10Y-minus-2Y term-premium slope; fitted exposure is removed before rebuilding the state with the baseline entry, release, and breadth rules. Cells report annualized coefficients and wild run-bootstrap p -values in brackets ($B = 9,999$). The stripped state has 125 months in 18 runs, versus 92 months in 15 runs for the baseline, and the two share 83 months. Of 1,134 inversion curve-months in the 1980:01–2026:02 slope panel, 942 (83 percent) remain inverted after stripping. This is a fitted adjustment for a common U.S. proxy, not a decomposition of foreign country-specific term premia. The high-beta effect remains statistically detectable; the carry coefficient attenuates and becomes imprecise.

The evidence therefore leaves two live readings. Synchronized curves can summarize coordinated revisions to expected policy and growth, or they can reveal common risk-bearing information embedded in term premia. The currency cross-section travels under the available adjustment; the carry magnitude does not. A consistent international expected-rate/term-premium panel is the data extension most likely to separate these interpretations.

6.3 Calendar, episode influence, and international breadth

Carry spot coefficients are negative in both calendar halves: −11.41 points before 2005 and −15.85 afterward. The S&P-beta portfolio is less stable, with −4.01 ($p = 0.287$) in the first half and −11.91 ($p = 0.010$) in the second. These are recursive historical stability facts, not a holdout, because the state was evaluated over the full history.

Leaving out one episode at a time keeps carry coefficients within roughly one-third of the full-sample value, and the largest reported beta-portfolio bootstrap p -value is 0.006. The exercise does not report every deletion coefficient or a joint leave-two-episode audit. The results are not reducible to one episode, but fifteen episodes remain the effective number of common-state realizations.

Seven floating emerging-market currencies add international breadth without constructing the G10 state. All seven bilateral signs are negative. The pooled spot

coefficient is -14.37 with $p = 0.104$, and the pooled total coefficient is -13.35 with $p = 0.079$. On the common 1996–2026 sample, active-state G10 and G10-plus-emerging-market spot means are similar, -12.79 and -12.51 percent. Total means differ, -8.67 and $+2.51$, because implied interest income is approximately 4.12 versus 15.02 percent. Shared spot repricing need not imply a universally negative total return when predetermined income differs; interpreting that difference as risk compensation requires additional structure.

6.4 An independently executable public 10Y–3M audit

The public audit rebuilds a nearby signal from current-vintage OECD monthly 10-year government yields and 3-month interbank rates distributed through FRED. It pairs that state with a BIS log spot-return proxy ranked on policy-rate differentials available by portfolio formation. The declared 64-rule family crosses raw versus live states, breadth one through four, release lengths one through three, one- and two-month return lags, and target leg sizes of two or three. All currencies tied at a cutoff enter with equal weight, so realized leg size can exceed its target. The code exhaustively reports every rule, but the family was not preregistered before the data were examined.

The baseline-like rule produces a -3.20 -point annualized log spot-return difference across 99 active months and eighteen episodes. Its inclusive circular-shift reference value is 0.335. Forty-one of 64 rules are negative, the baseline ranks 22nd from most negative to most positive, and its common-calendar maximum-standardized-coefficient reference value is 1.000. Negative signs are frequent, but the baseline estimate is imprecise and no rule meets the 5-percent family reference criterion.

Table 10: Independent public 10Y–3M proxy audit

Public-data specification	Log spot difference	Raw rotation ref.	Active months	Episodes
Baseline-like rule	−3.20	0.335	99	18
Exclude episodes containing 1998:09, 2008:10, or 2020:03	−0.46	0.826	88	15
First calendar half, 1988–2004	−5.73	0.382	49	9
Second calendar half, 2005–2025	−1.02	0.776	50	10
Delete five worst active months	+0.94	—	94	—

Notes: Annualized percentage-point active-minus-inactive differences in a BIS policy-ranked log spot-return proxy. Each leg targets three currencies; all currencies tied at a cutoff enter with equal weight, so realized leg size can exceed three. The baseline-like rule uses current-vintage OECD monthly 10-year government yields minus 3-month interbank rates distributed through FRED, fresh entry, breadth two, two consecutive increases for release, and a one-month return lag. It is distinct from the baseline 10Y–2Y state and is not an executable forward return. Across the complete declared family, 41 of 64 coefficients are negative, the baseline ranks 22nd from most negative to most positive, and the common-calendar max- $|z|$ rotation-reference value is 1.000. The crisis-deletion reference rotates the resulting complete-case sequence after excluded episodes are removed. The last row is outcome-conditioned and intentionally has neither a p -value nor an episode count, because deleting realized months mechanically splits state runs.

The public same-universe checks sharpen and limit that conclusion. Jointly deleting each currency from signal construction and the spot basket produces nine negative influence estimates, ranging from −5.08 when Australia is removed to −0.60 when the pound is removed. These estimates share most dates and are not nine independent tests. A geographically disjoint European-curve signal precedes a −15.84-point non-European basket difference across ten observed episodes (raw rotation-reference value 0.062); the reverse direction is −0.34 across nine episodes (reference value 0.887). These auxiliary references rotate each complete-case sequence; internal missing months are compressed rather than treated as literal calendar observations. The two designs also use different target leg sizes and coverage thresholds. Their asymmetry does not validate a universal common factor.

Concentration matters as well. Removing public-proxy episodes containing September 1998, October 2008, or March 2020 attenuates the baseline to −0.46 (complete-case rotation-reference value 0.826). Deleting the five worst active months changes the coefficient to +0.94. The public audit is a novel robustness result because its data, rule

grid, finite-rotation references, episode ledger, and state recursion are independently executable from the saved current-vintage snapshot. It disciplines the central claim sharply: a nearby public configuration often produces the same coefficient sign, but zero rules meet the 5-percent common-calendar family reference criterion and the baseline-like estimate disappears outside the largest losses.

6.5 What the design establishes

The estimates establish a large historical conditional spot-return shift and a nonlinear cross-country configuration. The S&P-based tests document an exposure ordering with the stated benchmark and inferential limits. The U.S., average-slope, own-curve, risk-control, and U.S.-term-premium comparisons narrow nearby explanations. The independent public family shows frequent negative coefficients but no rule meets its 5-percent common-calendar rotation-reference criterion.

Three boundaries remain decisive. The public audit does not regenerate the 10Y–2Y state, its fifteen-episode ledger, or the full specification search behind that state. Money-market differentials are not executable one-month forward returns after basis, spreads, formation timing, and financing costs. Finally, prediction and exposure ordering do not identify the shock or compensation friction. These boundaries define the next empirical agenda; they do not dilute the central economic result.

7 Why Can a Public State Forecast Poor Carry Returns?

7.1 Spot prediction, known income, and total returns

The empirical results separate three objects. First, the state predicts a spot depreciation of the high-rate currencies relative to the funding currencies. That fact is not itself a violation of uncovered interest parity: high-rate currencies can depreciate. Second, the portfolio earns a positive interest differential known at formation. Third, the money-market-differential total-return proxy is negative on average inside the state because the spot movement exceeds that income.

The income leg moves in the compensating direction: it averages 3.60 percent per year outside the state and 5.14 percent inside it, a +1.54-point difference with episode-bootstrap $p = 0.260$. That rise is too small to offset the -12.85 -point spot gap, leaving the -11.31 -point total-return gap. The differential is observed income, not identified risk compensation.

This accounting creates the pricing question. If a public state reveals bad-state covariance, a frictionless asset-pricing benchmark normally assigns compensation before the payoff. A negative conditional mean can still appear in a small sample of rare episodes, or when the measured total return differs from a feasible forward payoff. If it survives those explanations prospectively, prices or positions must adjust slowly enough for current expected losses to exceed compensation already embedded in the differential.

7.2 A benchmark restriction

Let an adverse event of size $J > 0$ occur at $t + 1$ with conditional probability p_t . Write the known income as $C_t = \bar{C}_t + \mu J \mathbb{E}_{t-1}[p_t]$, where $\bar{C}_t$ is ordinary carry and the second term represents compensation based on the earlier information set, with $\mu \geq 1$. Then

$$\mathbb{E}_t[rx_{t+1}] = \bar{C}_t + J \{ \mu \mathbb{E}_{t-1}[p_t] - p_t \}, \quad \mathbb{E}_t[rx_{t+1}] < 0 \iff p_t > \mu \mathbb{E}_{t-1}[p_t] + \frac{\bar{C}_t}{J}. \quad (10)$$

The inequality is the benchmark's only claim. Current perceived loss risk must rise enough to exceed both ordinary income and compensation based on prior beliefs. Fresh synchronized inversions can mark such an information change. The regression does not

observe p_t , estimate J , or identify the lag in compensation.

Three interpretations remain. The conditional mean may be compensation for adverse covariance estimated imprecisely from fifteen episodes. Portfolio adjustment or intermediary constraints may delay the currency response (Bacchetta and van Wincoop, 2010; Gabaix and Maggiori, 2015). Historical rule selection may also make the in-sample conditional mean more negative than its prospective value. These explanations imply different future evidence: executable forward payoffs, position or flow adjustments, and performance under a frozen state.

The paper does not choose among them. Its ambitious, testable implication is upstream: a cross-country bond-market configuration known before the payoff historically predicts losses on the persistent high-rate and high-beta currency exposures. A prospective episode under the dated algorithm can distinguish persistent pricing information from historical classification.

8 Conclusion

The cross-country configuration of yield curves predicts a distinct currency-return state. When fresh 10Y–2Y inversions are live in at least two non-U.S. G10 economies, next-month carry spot returns are 12.9 percentage points per year lower. The known interest differential absorbs only part of that depreciation. The opposite signs in the separate exactly-one and exactly-two indicator regressions, together with the portfolio horse race, distinguish the configuration from inversion in general, the U.S. curve, and the reported continuous average-slope and own-state measures. Those indicator coefficients are each measured against a different complement and are not a direct one-versus-two contrast.

The cross-section gives that forecast economic content. High-rate currencies lose relative to low-rate funding currencies, and bilateral losses grow with predetermined S&P equity beta. Persistent portfolio cores show that the result is not created solely by rotation at signal dates. The S&P-based exposure ordering is more stable across the reported tenor, release, and common-term-premium perturbations than the precise carry magnitude, although the nine-point fit is descriptive and the full-sample beta result depends on the equity benchmark.

An independently executable public 10Y–3M audit frequently produces a negative coefficient across its declared family, but no rule meets the 5-percent common-calendar rotation-reference criterion and the result attenuates after crisis-episode deletion. The public proxy therefore does not independently confirm the baseline relation. It shows why the exact carry coefficient cannot be treated as invariant to tenor, state persistence, or outcome construction. The main empirical object is the baseline configuration and the exposure-ordered loss it precedes.

The next decisive evidence is now well defined. Freeze the 10Y–2Y algorithm and vintage protocol, release the complete country-month state matrix, test each currency with its own curve excluded, and measure executable forward payoffs in the next unused episode. Those tests can determine whether the configuration is a durable international signal and whether its spot forecast translates into a negative conditional carry payoff.

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Online Appendix to

When Yield Curves Invert Together: Predicting Currency Carry Losses

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August 2026

Abstract

This appendix gives the complete signal algorithm and timing conventions, derives the return accounting and compensation benchmark, reports supplementary estimates and adverse robustness, and documents two reproducible public-data exercises. The first reveals a complete 64-rule family for an independently sourced 10Y–3M curve proxy; the second studies synchronized delivered easing and an exploratory country-combination screen. The exercises are kept distinct from the baseline 10Y–2Y evidence.

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A Signal construction and timing

This appendix gives the complete deterministic state transition used in the paper. It fixes the information set and separates fresh entry from confirmed release. It does not establish that the tenor, breadth threshold, and release length were selected independently of the return history.

A.1 Universe and information time

The nine non-dollar G10 currencies are the Australian dollar, Canadian dollar, Swiss franc, euro (Deutsche mark before 1999), British pound, Japanese yen, Norwegian krone, New Zealand dollar, and Swedish krona. The U.S. curve is excluded from the baseline count. For country i , define

$$q_{i,t} = y_{i,t}^{(10Y)} - y_{i,t}^{(2Y)}, \quad I_{i,t} = \mathbf{1}\{q_{i,t} < 0\}.$$

All yields are end-of-month observations. A state measured at the end of month t may condition only returns realized during month $t + 1$.

The return sample ends in 2026:02. Most money-market-rate series end in 2026:01 and therefore position the 2026:02 portfolio return under the $t \rightarrow t + 1$ convention. The Australian, Canadian, and Swedish rate series extend further and are truncated to the common formation endpoint. The final return month does not use same-month rate information.

A.2 Curve-level transition rule

Each curve carries a live indicator $L_{i,t}$, a consecutive-steepening counter $k_{i,t}$, and an eligibility flag. A fresh inversion observed at $t - 1$ is a crossing from $q_{i,t-2} \geq 0$ to $q_{i,t-1} < 0$; it makes the curve live at t . The update order is delayed entry, steepening-counter update for curves already live, release after the second consecutive increase, and finally cross-country aggregation.

Table 1: Curve-level state transition

Prior condition	Information used at t	End-of-month action
Eligible and not live at $t - 1$	$q_{i,t-2} \geq 0, q_{i,t-1} < 0$	Set $L_{i,t} = 1$; initialize $k_{i,t} = 0$
Live	$q_{i,t} > q_{i,t-1}$	Increase $k_{i,t}$ by one
Live	$q_{i,t} \leq q_{i,t-1}$	Reset $k_{i,t} = 0$
Live; counter reaches two	Any current slope sign	Set $L_{i,t} = 0$ after the second increase
Released but still inverted	$q_{i,t} < 0$	Remain inactive and ineligible
Released and non-inverted	$q_{i,t} \geq 0$	Become eligible for a future fresh crossing
Live	Current or lagged slope missing	Reset $k_{i,t} = 0$; do not release
Not live	Either side of the lagged crossing missing	Do not enter

A curve can therefore be released while still inverted. It cannot re-enter until it first becomes nonnegative and then crosses below zero again. An interrupted or unobserved steepening sequence resets the counter. These rules make the state depend on the path of slopes, not only on their current signs. They also impose two timing steps: a crossing at $t - 1$ makes the curve live at t , and the resulting state positions the return in $t + 1$.

A.3 Cross-country aggregation and raw benchmark

After updating all curves, form

$$N_t = \sum_{i=1}^9 L_{i,t}, \quad S_t = \mathbf{1}\{N_t \geq 2\}.$$

The baseline state S_t is active in 92 of 458 return months and forms fifteen contiguous episodes. The corresponding current-inversion benchmark is

$$R_t = \mathbf{1}\left\{\sum_{i=1}^9 I_{i,t} \geq 2\right\}.$$

The return in month $t + 1$ is paired with S_t or, in the benchmark, R_t . The raw state asks whether fresh entry and confirmed release add information beyond the current sign of the curves.

Table 2: Incidence of the synchronized-inversion state across eras

Era	Sample months	Active months	Active share (%)
1988–94	84	33	39.3
1995–2000	72	19	26.4
2001–07	84	17	20.2
2008–09	24	6	25.0
2010–19	120	3	2.5
2020–21	24	3	12.5
2022–26	50	11	22.0

Notes: Counts reproduce the baseline era accounting. They describe when the state occurs, not whether its conditional-return coefficient differs across eras.

A.4 Initialization and audit outputs

At the sample start, every curve is initialized as inactive. A curve can enter only after both sides of a fresh crossing are observed. Missing values neither trigger entry nor count toward release; a missing comparison breaks consecutiveness and resets the confirmation counter. The recursion produces, for every country-month, the slope, raw inversion, fresh-entry flag, live status, steepening counter, release flag, and eligibility status. It then produces N_t , S_t , and episode start, end, and duration. These fields are the minimum audit trail needed to distinguish crossing month, live-state month, and return month and to reproduce the fifteen-episode count.

The reported real-time audit recomputes classifications at 435 truncated month-ends and finds no changes in prior signal, count, or own-state classifications. Recomputing the conditioning series and sorts at 492 formation dates produces no portfolio-membership mismatches. One additional return lag attenuates the carry-spot coefficient from -12.9 to -6.3 points and the S&P-beta coefficient from -7.5 to -4.7 ; forward-shift placebos remove the adverse association. These checks support the implemented information clock, although they are not a substitute for a fully versioned vintage archive.

B Accounting and mechanism details

This appendix separates return accounting, an exposure-ordering device, and a candidate compensation friction. The first is definitional. The latter two organize the evidence but are not structurally identified by the predictive regressions.

B.1 Currency and portfolio returns

Let $s_{i,t}$ be the log dollar price of currency i and let $d_{i,t} = (i_{i,t} - i_{US,t})/12$. The paper defines the one-month return proxy and carry portfolio as

$$\begin{aligned} rx_{i,t+1} &= d_{i,t} + \Delta s_{i,t+1}, \\ rx_{t+1}^{\text{carry}} &= \frac{1}{3} \sum_{i \in \mathcal{H}_t} rx_{i,t+1} - \frac{1}{3} \sum_{i \in \mathcal{F}_t} rx_{i,t+1} \equiv C_t + X_{t+1}. \end{aligned} \quad (1)$$

The high-rate set is $\mathcal{H}_t$ and the funding set is $\mathcal{F}_t$. The construction makes $C_t > 0$ in every sample month, so the portfolio loses only when $X_{t+1} < -C_t$. The return proxy combines a simple annualized differential divided by twelve with a log spot change and is therefore a first-order monthly approximation, not an exact executable forward payoff. Empirically, C_t averages 3.9 percent per year and has a minimum of 0.7 percent. The active-state spot-return difference is -12.85 percentage points and the total-return difference -11.31 , so known income absorbs only part of the currency loss.

To organize the cross-section, write

$$\Delta s_{i,t+1} = -\lambda_i g_{t+1} + \eta_{i,t+1}, \quad (2)$$

where $g_{t+1} > 0$ is an adverse component shared across currencies, λ_i is a latent loading, and $\eta_{i,t+1}$ is idiosyncratic. A diversified long-short portfolio partly reduces idiosyncratic components; it retains the shared component when average λ_i differs between its two legs. The expanding-window equity beta $\hat{\beta}_{i,t}$ is a predetermined proxy for λ_i , not the same primitive parameter.

B.2 Yield slopes and synchronized information

A long-minus-short yield spread combines differences in expected short-rate averages with differences in maturity-specific term premia (Campbell and Shiller, 1991; Hamilton and Kim, 2002). Consequently, one inversion is consistent with expected easing but does not reveal whether the news is domestic, global, expectations-driven, or a term-premium movement. Synchronization raises the plausibility of shared information because several markets cross a salient boundary, but it does not identify the shared shock.

The exact-one versus exact-two estimates give this idea empirical content. Exactly one live inversion is associated with carry spot and total coefficients of 8.33 and 8.78 percentage points

per year. At exactly two, the coefficients are -15.72 and -15.13 . The sign reversal indicates nonlinearity in cross-country breadth within the sample. It does not establish two as an optimal population threshold.

B.3 Delayed compensation as a candidate friction

Let A_{t+1} indicate an adverse event that reduces carry payoff by $J > 0$. Write

$$rx_{t+1}^{\text{carry}} = C_t - JA_{t+1} + \xi_{t+1}, \quad \mathbb{P}_t(A_{t+1} = 1) = p_t, \quad \mathbb{E}_t[\xi_{t+1}] = 0. \quad (3)$$

Then $\mathbb{E}_t[rx_{t+1}^{\text{carry}}] = C_t - p_t J$. For a tradable excess return in a frictionless benchmark, the Euler condition $\mathbb{E}_t[m_{t+1} rx_{t+1}^{\text{carry}}] = 0$ requires compensation when losses occur in high-marginal-utility states. Publicly visible risk therefore does not automatically imply a negative conditional mean.

The candidate friction places an information lag in compensation. Suppose

$$C_t = \bar{C}_t + \mu J \mathbb{E}_{t-1}[p_t], \quad \mu \geq 1, \quad (4)$$

where $\bar{C}_t$ is ordinary carry income and the second component prices the current-period hazard using the earlier information set. Combining equations (3) and (4) gives

$$\mathbb{E}_t[rx_{t+1}^{\text{carry}}] = \bar{C}_t + J \{ \mu \mathbb{E}_{t-1}[p_t] - p_t \}. \quad (5)$$

Hence

$$\mathbb{E}_t[rx_{t+1}^{\text{carry}}] < 0 \iff p_t > \mu \mathbb{E}_{t-1}[p_t] + \frac{\bar{C}_t}{J}. \quad (6)$$

The inequality is the model's economic content. A rise in perceived risk is insufficient: the current expected loss must exceed both previously embedded event compensation and ordinary carry income. Fresh synchronized inversions may mark such a belief innovation, but the paper does not observe p_t or identify the adjustment friction.

B.4 Conditional uncovered interest parity

The supporting Fama regression estimates the appreciation slope separately by state:

$$\Delta s_{i,t+1} = a_i^{(s)} + \phi_s d_{i,t} + u_{i,t+1}, \quad S_t = s, \quad s \in \{0, 1\}. \quad (7)$$

Under the paper's quote convention, uncovered interest parity implies $\phi_s = -1$. The pooled estimate is 0.50 outside the state and -1.33 inside it. The point estimates say that high-rate currencies depreciate in the state in which carry loses. The state interaction has a Newey–West t -statistic of -2.39 but an episode-bootstrap p -value of 0.340. The pooled panel contains many currency-months, whereas the common state changes across only fifteen episodes; the episode result is therefore the appropriate warning against treating -1.33 as a precise structural slope.

C Supplementary evidence and robustness

All exercises in this appendix use the same historical sample on which the state was assessed. They measure sensitivity and economic breadth; they do not create a frozen out-of-sample test.

C.1 Distribution and episode influence

Removing each portfolio’s full-sample bottom-decile months leaves negative carry spot and total state effects. Because the deletion conditions on the realized outcome, it diagnoses tail concentration rather than estimating a preferred conditional mean.

Table 3: State effects after removing bottom-decile months

Portfolio component	Remaining signal months	Difference (%/yr)	Episode wild p
Carry spot	77	-7.24	0.006
Carry total	77	-5.50	0.024
High-beta spot	77	-3.72	0.084
High-beta total	78	-3.68	0.060

Notes: Each regression drops that return series’ full-sample bottom-decile months from both states. The exercise asks whether a few crash observations mechanically generate the conditional-mean result; it does not address selection among alternative signal definitions. Wild-bootstrap p -values cluster by IYC episode.

The median and every reported lower quantile also move left. For carry spot returns, the monthly median changes from 0.35 percent outside the state to -0.76 percent inside, the 25th percentile from -0.93 to -1.98 , and the 10th percentile from -2.35 to -3.94 . This evidence is more informative about a broad distributional shift than the fact that all five worst months occur under the lagged state.

The beta-sorted portfolio exhibits a different distributional pattern. Its bottom-decile frequency is 16.3 percent in active months and 8.5 percent outside them ($p = 0.046$); at a fixed -4 -percent monthly cutoff, the corresponding rates are 9.8 and 4.6 percent ($p = 0.033$). Annualized volatility is nearly unchanged, 7.94 versus 7.75 percent, with a variance ratio of 1.05 ($p = 0.80$), and reported loss severity conditional on crossing the tail cutoff is similar across states. These outcome-defined diagnostics describe a higher realized tail frequency, not an identified change in disaster probability.

Leave-one-episode-out carry estimates remain within roughly one-third of the full-sample coefficient. The high-beta effect remains statistically detectable under every one-episode deletion, with a largest reported bootstrap p -value of 0.006. These checks reduce dependence on a single episode but do not overcome the limited number of episodes.

C.2 Calendar stability

The carry-spot coefficient is negative in both calendar halves. The high-beta result is less stable and is concentrated after 2005. The split is descriptive because the rule was not frozen on the first half before evaluation on the second.

Table 4: The headline estimates in two halves of the sample

Series	1988:01–2004:12			2005:01–2026:02		
	b	$t_{episode}$	wild p	b	$t_{episode}$	wild p
Carry spot	-11.41	-2.54	0.037	-15.85	-1.99	0.010
Carry total	-10.20	-2.04	0.052	-14.76	-1.84	0.047
High-beta spot	-4.01	-1.12	0.287	-11.91	-4.07	0.010
High-beta total	-3.03	-0.80	0.448	-11.54	-3.85	0.011
Months	204			254		
Signal months	54			38		
Episodes	7			8		

Notes: Each column reports the state regression $y_t = a + b\mathbf{1}\{\text{IYC}\}_{t-1} + e_t$ estimated separately by calendar half. Coefficients are annualized percentage points; $t_{episode}$ clusters by IYC episode, and wild-bootstrap p -values use $B = 9,999$. This is a stability check, not a frozen out-of-sample test.

C.3 State-definition sensitivity

The reported grids vary breadth, release confirmation, and the short maturity. Separate exactly-one and exactly-two indicator regressions have coefficients with opposite signs; they are not a direct contrast because each indicator is compared with a different complement. At-least-three and at-least-four rules have fewer active months and later coverage. Releasing after one steepening month is noisier; requiring three retains the state longer. Rebuilding the state with a 10Y–3M rather than 10Y–2Y slope preserves the broad adverse ordering but changes episode timing.

The raw current-inversion state is active in 165 months, compared with 92 for the live state. Relative to the raw count, the live state improves tail concentration rather than absolute capture at every cutoff. The raw count contains 9 of the 23 worst-5-percent months and 19 of the 46 worst-10-percent months, versus 8 and 15 for the live state. Because the raw count is active in 36 percent of the sample rather than 20 percent, its concentration ratios are lower—1.09 versus 1.73 at the 5-percent cutoff and 1.15 versus 1.62 at the 10-percent cutoff. Only in the extreme five-month tail does the live state dominate absolute capture, 5 of 5 versus 2 of 5. The latch can retain losses that arrive after curves begin to re-steepen, but its extreme-tail concentration is partly mechanical because the historical crash chronology helped motivate persistence.

Breadth above two mainly measures episode maturity rather than a monotone risk dose. Months with at least three live curves have median episode age six months, compared with three months for exactly-two months, and eleven of fifteen episodes begin at exactly two. Conditional on the baseline state, a three-or-more indicator adds no reported information with or without episode-age controls ($p \geq 0.27$). The exact-two contrast is therefore an entry/configuration result, not evidence that each additional inversion raises loss risk.

A reciprocal-age version of the current-inversion count separates two timing margins. It has a high-beta spot coefficient of -14.87 ($p = 0.001$), larger than the baseline live state’s -7.53 , but a smaller carry coefficient of -10.43 ($p = 0.040$) than the baseline’s -12.85 . Under the full control set, the age-weighted high-beta coefficient remains -13.82 ($p = 0.008$), whereas its carry coefficient

attenuates to -7.55 ($p = 0.084$). These in-sample comparisons suggest that exposure repricing is concentrated near fresh information, while the path-dependent release rule is more important for carry losses that continue into re-steepening.

Table 5: Measurement perturbations and exposure ordering

State construction	Carry spot	Wild p	High-beta spot	Wild p
10Y-2Y baseline	-12.85	0.001	-7.53	0.010
10Y-3M slope	-4.68	0.146	-5.25	0.011
U.S.-term-premium-adjusted 10Y-2Y	-5.42	0.107	-6.17	0.004

Notes: Coefficients are annualized percentage-point active-state differences. The 10Y-3M row rebuilds every curve state using a different short maturity. The term-premium row removes each national slope’s fitted exposure to the U.S. ACM 10Y-minus-2Y slope premium before rebuilding the state. Both perturbations attenuate carry while retaining the beta-sorted exposure result.

Table 6: State-definition multiverse: reported and untested dimensions

Dimension	Reported neighborhood	Remaining uncertainty
Breadth	At least 1-4; exactly 1-2	Frequency-matched and country-combination rules
Release	One-, two-, and three-month confirmation	Other persistence and duration-matched placebo rules
Tenor	10Y-2Y baseline; 10Y-3M check	Other maturities and inversion cutoffs
Freshness	Fresh crossing versus current inversion	Alternative age decay and entry delay
Geography	Non-U.S. G10 baseline; U.S.-inclusion checks	Other developed-market sets and country combinations
Portfolio	Three-by-three carry and beta sorts	Other leg sizes, volatility scaling, feasible forwards
Timing	Month-end recursion and one extra lag	Vendor timestamps, asynchronous closes, vintage data

C.4 Bilateral cluster and own-curve specifications

The full country-level horse race is reported below. The cluster includes the focal country’s curve in every row. These regressions ask whether the configuration remains informative after controlling for the currency’s own state and slope variables; they do not show that other countries’ curves alone predict the focal currency.

Table 7: Bilateral cluster and own-curve specifications

Currency	Own-state months	Own state alone $b(t_{NW}) [p]$		Cluster alone $b(t_{NW}) [p]$	
AU	47	−8.76	(−1.65) [0.044]	−13.21	(−1.83) [0.015]
CA	40	+0.71	(+0.19) [0.858]	−4.61	(−1.33) [0.203]
CH	12	−20.12	(−1.91) [0.303]	−0.89	(−0.16) [0.867]
EU	25	+5.26	(+0.62) [0.589]	−1.19	(−0.20) [0.831]
GB	59	−3.84	(−0.59) [0.585]	−3.00	(−0.63) [0.542]
JP	22	+4.58	(+1.03) [0.298]	+2.35	(+0.58) [0.565]
NO	82	+4.53	(+0.84) [0.415]	−3.96	(−0.71) [0.520]
NZ	48	−11.42	(−1.33) [0.282]	−13.28	(−1.99) [0.011]
SE	13	−13.05	(−1.02) [0.497]	−4.94	(−0.76) [0.416]
Pooled (currency FE)	—	—	—	−4.74	(−2.34) [— — —]

Currency	Cluster, with own state $b(t_{NW}) [p]$	Own state, with cluster $b(t_{NW}) [p]$	Full-spec. cluster $b(t_{NW}) [p]$	Own-curve joint Wald p
AU	−14.18 (−1.38) [0.133]	+2.19 (+0.23) [0.859]	−12.92 (−1.31) [0.262]	0.408
CA	−6.01 (−1.45) [0.207]	+4.50 (+1.04) [0.356]	−5.55 (−1.46) [0.282]	0.801
CH	+1.28 (+0.21) [0.848]	−20.95 (−1.81) [0.272]	+1.15 (+0.18) [0.875]	0.115
EU	−3.40 (−0.51) [0.584]	+8.13 (+0.80) [0.552]	−4.00 (−0.61) [0.560]	0.488
GB	−2.00 (−0.45) [0.622]	−2.82 (−0.44) [0.668]	−0.44 (−0.10) [0.928]	0.486
JP	+1.68 (+0.38) [0.733]	+3.41 (+0.71) [0.413]	+6.37 (+1.32) [0.251]	0.006
NO	−6.85 (−1.15) [0.289]	+7.43 (+1.42) [0.221]	−8.17 (−1.23) [0.281]	0.233
NZ	−12.00 (−2.06) [0.011]	−3.22 (−0.41) [0.748]	−8.56 (−1.26) [0.220]	0.373
SE	−3.73 (−0.61) [0.526]	−10.58 (−0.92) [0.540]	−2.99 (−0.48) [0.662]	0.150
Pooled (currency FE)	−4.74 (−2.34) [— — —]	−0.01 (−0.01) [— — —]	—	—

Notes: Spot returns against the U.S. dollar, 1988:01–2026:02. Each compact cell reports the annualized coefficient, Newey–West (12) t -statistic in parentheses, and wild episode-bootstrap p -value in brackets. The full specification adds the currency’s own 10Y–2Y slope, its one-month change, its slope relative to the U.S. curve, and the average G10 slope and change. The final column is the Newey–West Wald p -value for the joint null on that currency’s own state, relative slope, and relative level conditional on the cluster. The pooled row reports currency fixed effects and no bootstrap inference. The cluster used in every row includes the focal country’s curve. This table conditions on own-curve information; it is not a leave-one-country-out signal.

C.5 Exposure breadth and composition

The carry and beta portfolios have persistent cores rather than changing membership only when the state turns on. New Zealand occupies the high-rate leg in 85 percent of all months and 91 percent of state months; Australia in 71 and 73 percent. Japan and Switzerland remain the principal funding currencies. There is also rotation around the core: Norway’s carry-long share falls from 62 to 34 percent in state months, while the pound’s rises from 43 to 70 percent. The return result is therefore not created solely by state-month resorting, but portfolio composition is not fixed.

Table 8: Complete portfolio composition in the full sample and active state

Currency	Long share		Funding share	
	Full	State	Full	State
<i>Panel A: carry portfolio</i>				
New Zealand	85	91	2	1
Australia	71	73	4	0
Norway	62	34	3	5
United Kingdom	43	70	4	0
Canada	20	21	23	14
Sweden	15	12	33	36
Euro area	4	0	51	47
Japan	0	0	83	97
Switzerland	0	0	96	100
<i>Panel B: equity-beta portfolio</i>				
Canada	100	100	0	0
Australia	97	92	0	0
New Zealand	77	76	0	0
Norway	26	32	0	0
Euro area	0	0	46	45
United Kingdom	0	0	52	47
Sweden	0	0	34	59
Japan	0	0	73	65
Switzerland	0	0	95	85

Notes: Entries are percentages of months in which each currency appears in the indicated three-currency leg. Full denotes all 458 portfolio months; State denotes the 92 active-state months. Column sums can differ slightly from 300 because the displayed shares are rounded to integer percentages.

The seven floating emerging-market currencies are not used to construct the G10 state. All seven bilateral state coefficients are negative; only the rand is individually detected at conventional levels (-14.68 , $p = 0.015$), while the rupee is marginal (-7.98 , $p = 0.061$) and the other five estimates are imprecise. Pooled spot and total effects are -14.37 and -13.35 percentage points per year, with episode-bootstrap p -values of 0.104 and 0.079. In the same EM panel, source-reported pooled spot coefficients are -4.54 for the raw current count and -3.92 for its six-month age filter. This ordering favors the live construction in sample, but all three designs use the same global dates and the live estimate itself is imprecise at the episode level.

The common-sample comparison separates international spot repricing from compensation.

Table 9: Spot repricing and compensation across currency universes, 1996–2026

Universe	Return leg	Inactive mean	Active mean	State difference	Episode wild p
G10	Spot	2.96	−12.79	−15.75	< 0.001
G10	Total	6.22	−8.67	−14.89	0.003
G10+EM	Spot	0.46	−12.51	−12.97	0.176
G10+EM	Total	11.68	2.51	−9.18	0.235

Notes: Annualized percentages over the common 1996:01–2026:02 sample. The G10+EM universe adds seven floating emerging-market currencies and applies the same three-by-three rate sort. Active-state income implied by total minus spot is approximately 4.12 percent for G10 and 15.02 percent for G10+EM. The nearly equal spot means but different total means show how predetermined carry income changes the payoff from similar currency repricing.

The combined sort is economically concentrated. Emerging-market currencies occupy 99 percent of carry-long slots and 93 percent of high-beta-long slots; the Brazilian real alone enters the carry-long leg in 93 percent of months. The comparison is therefore close to a persistent emerging-market exposure against the yen–franc–euro funding bloc, not a broadly rotating sixteen-currency portfolio.

Baseline exposure tests use expanding S&P 500 betas. An oil-beta sort adds little independent variation because its return is largely spanned by the carry and equity-beta portfolios.

C.6 Conditional UIP and macro timing

Appendix B gives the conditional Fama specification. The pooled slope is 0.50 outside the state and −1.33 inside, but the state interaction has episode-bootstrap $p = 0.340$. The point estimate is consistent with high-rate currencies depreciating during the state, while the interaction remains imprecise at the episode level.

The carry-spot response is front-loaded. In the baseline projection it falls about 1.1 percent in the first month ($p < 0.01$) and 2.4 percent cumulatively by month four ($p = 0.06$). Onset-only and lag-controlled designs reach roughly −3.7 to −4.2 percent around months four through six, followed by a partial rebound. Because horizons overlap and the state is persistent, these are descriptive timing associations rather than causal impulse responses.

The leading indicator declines by about 0.6 percent over six months, with similar magnitudes around episode onsets and under lag controls. Industrial production declines about 1.8 percent at six months in the baseline projection but becomes imprecise with the fullest controls. The average G10 money-market rate is approximately flat for the following year across the baseline, onset, and controlled projections. The leading-indicator response is persistent across these specifications; the industrial-production response and a uniform coordinated-easing path are not.

C.7 Benchmarks, controls, and term premia

The U.S. inversion, average G10 slope, and its change do not span the synchronized state. Lagged NFCI and VIX controls also leave the carry coefficient near its baseline magnitude.

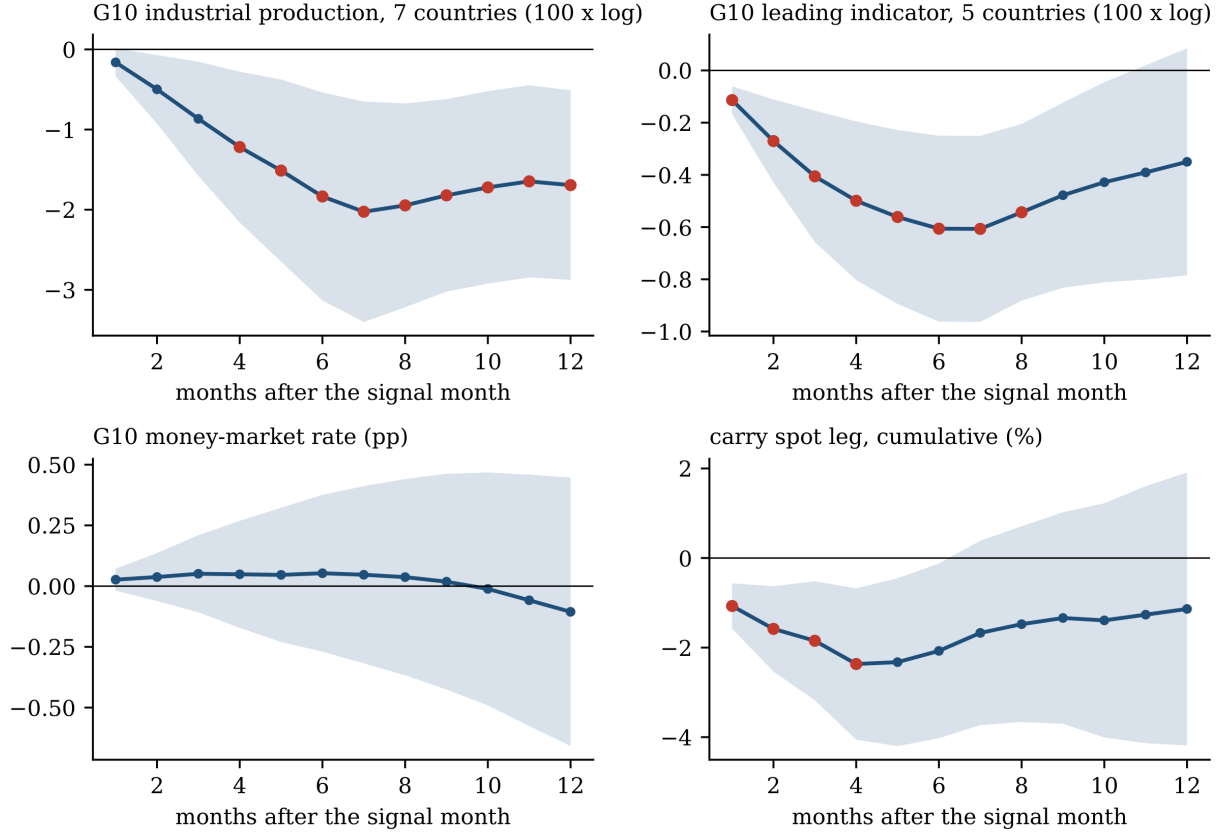


Figure 1: Outcomes following a synchronized-inversion state. The panels show local projections of cumulative changes in G10 industrial production, the OECD composite leading indicator, the average G10 money-market rate, and the carry spot leg. Bands are 90-percent Newey–West intervals; markers denote episode-bootstrap $p < 0.10$. Overlapping horizons and persistent states make these supporting temporal associations rather than causal responses.

Contemporaneous dollar and FX-volatility controls show that the realized loss is not simply the average dollar move or the measured volatility innovation, but they are not predictive controls.

Substituting G10 implied-volatility innovations preserves carry coefficients but weakens high-beta inference: high-beta point estimates remain near -7.5 percentage points and have p -values around 0.17 – 0.18 . The raw current-inversion count is absorbed by the full NFCI/VIX set, with carry $p \geq 0.58$ and high-beta $p \geq 0.12$. These adverse cells prevent the control evidence from being read as uniform across volatility measures or state constructions.

The term-premium adjustment is asymmetric. After removing fitted exposure to the U.S. ACM 10Y-minus-2Y term-premium slope, 83 percent of national inversions remain. The rebuilt state still contains all five worst carry months, but it expands from 92 to 125 active months. The high-beta coefficient is -6.2 percentage points with $p = 0.004$, while the carry coefficient is about -5.4 with $p = 0.11$. The smaller carry estimate reflects a changed classifier—and possible dilution from the additional months—as well as removal of fitted exposure to the U.S. premium. The exercise does not identify what share of the baseline association is attributable to a common term-premium component. Comparable foreign-country decompositions are unavailable.

C.8 Interpretive boundary

The conditional return gap, distributional shift, and exposure ordering are statistically detectable in the reported specifications. The conditional-UIP interaction and industrial-production response are imprecise; emerging-market breadth and the downcurve split are exploratory. Feasible forward returns, complete data-vintage auditing, a comprehensive state multiverse, and a frozen future sample remain untested.

D Independent public-data checks

This appendix reports two executable exercises. The first constructs an independently sourced 10Y–3M yield-curve state and reveals its complete 64-rule family. The second examines outcomes around synchronized delivered policy easing. The yield proxy is a nearby measurement audit; the easing event occurs downstream of the predictive state. Neither is substituted for the baseline 10Y–2Y analysis.

D.1 A public 10Y–3M yield-curve proxy

D.1.1 Inputs, state, and outcome

The public curve panel uses current-vintage OECD monthly observations distributed through FRED. For each of the nine currencies, the slope is a 10-year government yield minus a 3-month interbank rate. The euro series uses Germany through December 1998 and the euro-area aggregate from January 1999. At least six observed curves are required to classify a month. Swiss and Japanese short-rate coverage begins later than the main sample, so the early state uses the available cross-section. Current vintages, the short-rate concept, the observation convention, and the euro splice all differ from the baseline end-of-month 10Y–2Y panel.

The live recursion uses the same timing sequence as Appendix A: a crossing observed at $t - 1$ becomes live at t , a curve-specific counter records consecutive observed increases, release follows the required number of increases, and the country states are aggregated. A missing slope comparison resets the counter. The baseline-like public rule requires two live curves and two increases for release. The public outcome is a BIS policy-ranked log spot-return proxy, signed as foreign-currency appreciation against the dollar. For a one-month-lag rule, policy-rate differentials available by month t form the basket whose spot change is measured in $t + 1$. For a two-month-lag rule, the basket is re-sorted using rates available by $t + 1$ and its spot change is measured in $t + 2$; it is not a month- t portfolio held for two months. The target is three currencies on each side. All currencies tied at a cutoff receive equal weight, so a realized leg can contain more than three currencies; an outcome is missing unless every selected nonzero-weight currency return is observed. The proxy is not a money-market or forward excess return.

Every country-month is retained in a curve-recursion ledger with the provider series identifiers, fresh-entry flag, live status, counter, release, eligibility, inversion, and observation status. A monthly ledger records observed curves, current and live counts, state, next-month outcome, onset, and episode. These files make the public state reproducible without discretionary event dating.

D.1.2 Declared specification family and inference

The declared family contains 64 rules. Live states cross four breadth thresholds, three release lengths, two return lags, and two portfolio leg sizes, giving 48 rules. Raw current-inversion states cross four thresholds, two lags, and two leg sizes, giving 16 more. The code reports every declared

rule, but the declaration was not preregistered before the data were examined.

All family rules use the literal common complete-case calendar needed by both return horizons. For each rule, the reference distribution applies every common calendar displacement to the complete state sequence, includes the observed assignment, and doubles the smaller inclusive tail rank. The family reference standardizes each rule’s rotation distribution and records the maximum absolute standardized coefficient across all 64 rules at each identical displacement. Under the maintained null that simultaneous cyclic shifts of the state sequences relative to returns are exchangeable, the resulting comparison controls the declared family. Without that assumption it is a finite common-calendar rotation-reference diagnostic, not an unconditional randomization probability.

Table 10: Independent public 10Y–3M proxy audit

Public-data specification	Log spot difference	Raw rotation ref.	Active months	Episodes
Baseline-like rule	−3.20	0.335	99	18
Exclude episodes containing 1998:09, 2008:10, or 2020:03	−0.46	0.826	88	15
First calendar half, 1988–2004	−5.73	0.382	49	9
Second calendar half, 2005–2025	−1.02	0.776	50	10
Delete five worst active months	+0.94	—	94	—

Notes: Annualized percentage-point active-minus-inactive differences in a BIS policy-ranked log spot-return proxy. Each leg targets three currencies; all currencies tied at a cutoff enter with equal weight, so realized leg size can exceed three. The baseline-like rule uses current-vintage OECD monthly 10-year government yields minus 3-month interbank rates distributed through FRED, fresh entry, breadth two, two consecutive increases for release, and a one-month return lag. It is distinct from the baseline 10Y–2Y state and is not an executable forward return. Across the complete declared family, 41 of 64 coefficients are negative, the baseline ranks 22nd from most negative to most positive, and the common-calendar max- $|z|$ rotation-reference value is 1.000. The crisis-deletion reference rotates the resulting complete-case sequence after excluded episodes are removed. The last row is outcome-conditioned and intentionally has neither a p -value nor an episode count, because deleting realized months mechanically splits state runs.

The baseline-like estimate is −3.20 percentage points per year, with raw rotation-reference value 0.335 and common-calendar maximum-standardized-coefficient reference value 1.000. Forty-one of 64 rules are negative; none meets the 5-percent family reference criterion. Figure 2 orders rules by active-state frequency and then their declared fields, not by the estimated return. The baseline-like rule ranks 22nd from most negative to most positive.

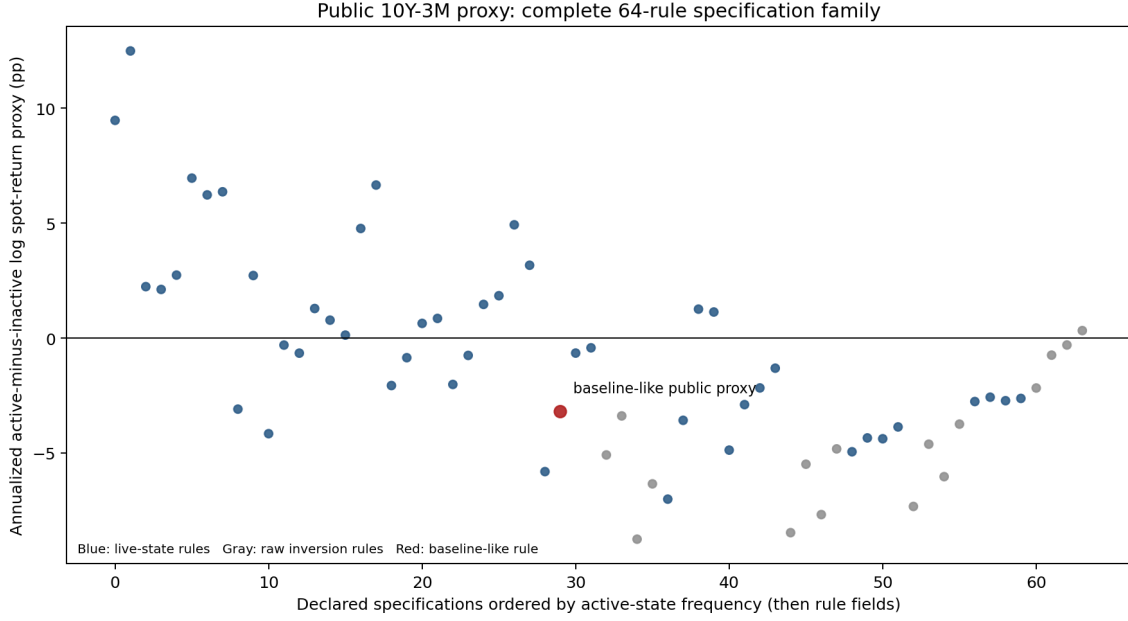


Figure 2: Complete public 10Y–3M rule family. Blue points are live states, gray points raw current-inversion states, and the red point is the baseline-like rule. Specifications are ordered by active-state frequency and fixed rule fields.

D.1.3 Exclusions, disjoint signals, and episodes

Removing each currency from both the public curve state and the outcome produces nine negative estimates. The samples and calendars overlap, so the signs are a sensitivity pattern rather than nine independent successes. For these influence and disjoint checks, the auxiliary reference rotates each complete-case sequence after dropping missing observations; it therefore compresses internal calendar gaps. The two geographically disjoint constructions produce different results: European curves precede a negative non-European basket difference, whereas non-European curves do not precede losses in the European basket.

Table 11: Public proxy exclusions and disjoint signals

Excluded from curve state and outcome	Log spot difference	Raw rotation ref.	Active months	Episodes
AUD	−5.08	0.133	69	16
CAD	−1.96	0.554	92	17
CHF	−4.51	0.204	92	17
EUR	−2.90	0.391	88	18
GBP	−0.60	0.827	88	16
JPY	−2.33	0.404	98	18
NOK	−1.32	0.655	70	14
NZD	−1.93	0.538	82	14
SEK	−1.95	0.591	91	16
<i>Geographically disjoint curve state and outcome</i>				
European curves → AUD/CAD/JPY/NZD	−15.84	0.062	38	10
Non-European curves → CHF/EUR/GBP/NOK/SEK	−0.34	0.887	44	9

Notes: Annualized active-minus-inactive differences in the log spot-return proxy. Each joint deletion removes a currency from both the public curve state and the public spot basket and requires five of the remaining eight curves. The resulting influence estimates overlap heavily and are not independent tests. Disjoint rows use no curve from a currency in the outcome basket. The European sensor requires four of five curves and targets one currency per side over four outcomes; the non-European sensor requires three of four curves and targets two per side over five. All cutoff ties are included and equally weighted. Reference values use inclusive circular shifts of each complete-case sequence after missing observations are dropped, compressing internal calendar gaps. They are not adjusted as an eleven-test family.

The public state produces eighteen episodes. Their cumulative outcomes are heterogeneous, with large negative contributions from the 1998 and 2008 episodes and several positive episodes. Removing episodes containing September 1998, October 2008, or March 2020 attenuates the baseline difference from −3.20 to −0.46 points. Deleting the five worst active months changes it to +0.94. The complete episode ledger makes this concentration inspectable.

Table 12: Public 10Y–3M proxy episode ledger

Episode	Onset	End	Months	Sum of log spot proxy	Worst month
1	1988:08	1989:12	17	−6.10	−4.62
2	1990:12	1991:05	6	−0.87	−1.15
3	1993:02	1993:02	1	−0.48	−0.48
4	1993:04	1993:05	2	−3.21	−3.05
5	1997:09	1997:09	1	−1.64	−1.64
6	1998:09	1999:01	5	−7.79	−7.04
7	1999:11	1999:12	2	+2.24	+0.26
8	2000:05	2001:02	10	+0.45	−2.31
9	2004:08	2005:07	12	+1.90	−1.61
10	2007:09	2008:04	8	−3.92	−3.66
11	2008:09	2008:10	2	−14.65	−11.66
12	2011:09	2011:12	4	+4.97	−2.47
13	2015:02	2015:05	4	−6.15	−4.03
14	2016:08	2016:09	2	+2.62	+0.96
15	2019:06	2019:11	6	−0.08	−2.95
16	2020:02	2020:05	4	−0.57	−6.08
17	2022:12	2023:09	10	+3.39	−1.59
18	2024:01	2024:03	3	+3.68	+0.34

Notes: Episodes use the baseline-like public 10Y–3M live state. The reported episode total is the sum of monthly percentage-point observations of the next-month BIS policy-ranked log spot-return proxy. Each side targets three currencies, with all cutoff ties included and equally weighted. The public 18-episode ledger is specific to this exercise and is distinct from the baseline 15-episode 10Y–2Y state.

The public audit produces an ambitious but bounded result. A nearby cross-country curve configuration yields negative coefficients in a majority of declared rules and in every joint delete-one influence check, yet the common-calendar family reference and crisis deletion remove conventional statistical detection. The exercise exposes measurement sensitivity using an independently executable current-vintage snapshot.

D.2 Delivered-easing mechanism proxy

The second exercise uses public data to examine outcomes around synchronized delivered policy easing. The event is intentionally later than the yield-curve state: inversions can encode expected easing, whereas policy-rate cuts record policymakers’ actions. It tests selected mechanism implications but does not validate the inversion classifier.

D.2.1 Event construction

The sample is January 1988 through December 2025. Let $r_{i,t}^{pol}$ be country i ’s BIS end-of-period policy rate and define

$$K_{i,t} = \mathbf{1}\{\Delta r_{i,t}^{pol} \leq -0.10\}, \quad B_t = \sum_i K_{i,t}, \quad (8)$$

where rates and the 10-basis-point cutoff are in percentage points. Let $D_t = 1$ when $B_t \geq 3$ and at least six country rates are observed. The onset indicator is

$$O_t = D_t \prod_{\ell=1}^3 (1 - D_{t-\ell}). \quad (9)$$

Thus an onset requires at least three cutting countries after three months without another synchronized-cut month. The rule produces fifteen onsets. These fifteen public easing onsets are not the paper’s fifteen inversion episodes; the identical counts are coincidental.

The public currency outcome is a spot-only proxy. BIS exchange rates are signed so that a positive return is foreign-currency appreciation against the dollar. At $t-1$, currencies are ranked by their policy-rate differentials against the United States; the proxy is long the three highest-ranked and short the three lowest-ranked. The rank is predetermined, but O_t is observed during month t , so month- t returns are contemporaneous event associations rather than feasible predictions made before easing.

Additional outcomes are CFTC non-commercial net positions scaled by open interest, the New York Fed ACM 10-year expected-rate and term-premium components (Adrian et al., 2013), the OECD G7 composite leading indicator, NFCI, and VIX. CFTC coverage is limited to seven directly matchable contracts and uses a documented Deutsche-mark/euro convention. ACM is a U.S. diagnostic, not a decomposition of the nine foreign slopes. The macro and financial-condition series are current vintage.

D.2.2 Event estimands and inference

The code uses two estimands. For a level outcome Y , the horizon- h event response is

$$\hat{\theta}_h^{\text{change}} = \frac{1}{n_h} \sum_{j \in \mathcal{J}_h} (Y_{\tau_j+h} - Y_{\tau_j-1}). \quad (10)$$

For a monthly return flow R , it is

$$\hat{\theta}_h^{\text{return}} = \frac{1}{n_h} \sum_{j \in \mathcal{J}_h} \sum_{\ell=0}^h R_{\tau_j+\ell}, \quad (11)$$

where τ_j indexes valid onset dates. Ninety-percent intervals resample onset episodes; leave-one-onset ranges diagnose influence. The reference distribution enumerates circular shifts of the complete onset sequence, retains shifts with the observed number of valid event outcomes, includes the observed assignment, and doubles the smaller inclusive-tail rank. With irregular missingness, the retained shifts need not form a transformation group; the resulting values are conditional finite-rotation references, not exact causal randomization tests.

Six declared primary outcomes receive Holm adjustment of their finite-rotation reference values: the shadow-carry spot return over months 0–1; the CFTC positioning change through month 3; ACM expected-rate and term-premium changes through month 3; the CLI change through month 6; and the NFCI change through month 3. The declaration is recorded in the analysis configuration and run manifest, but not in an immutable pre-analysis registration. VIX and other horizons are descriptive.

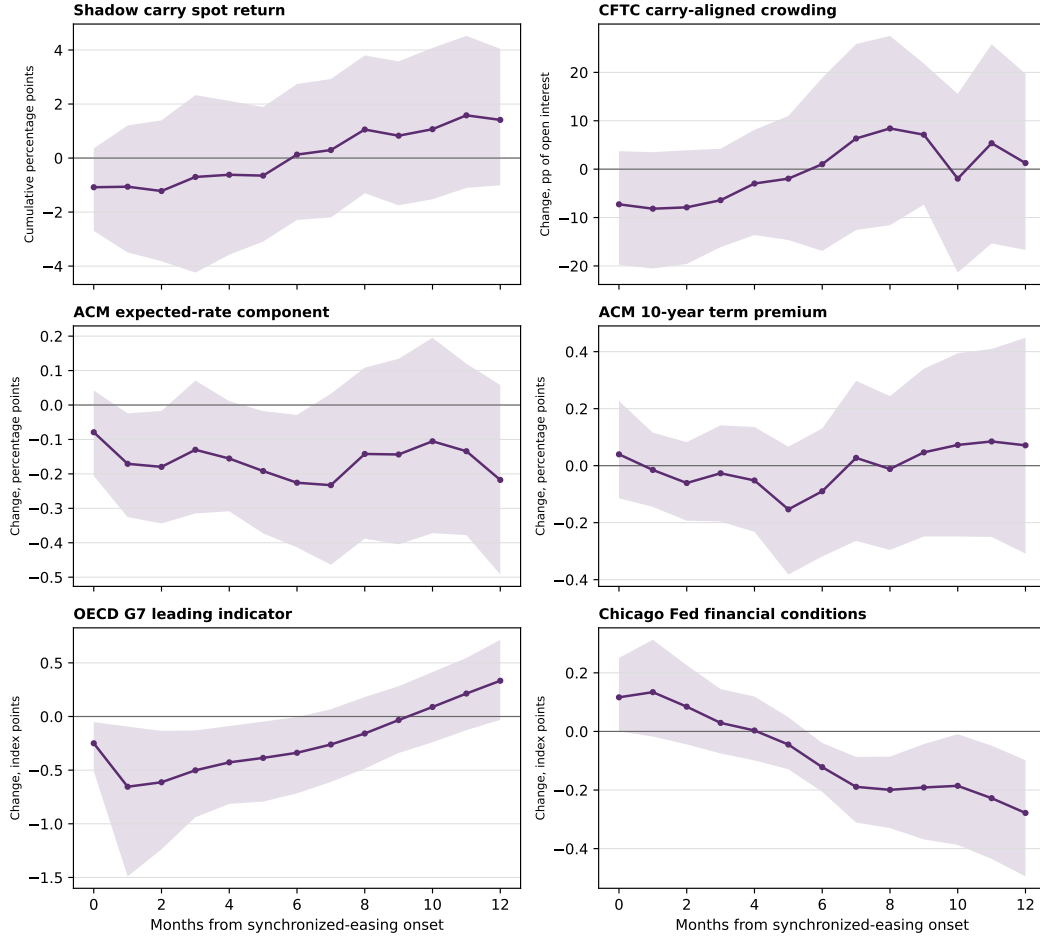
Table 13: Outcomes around synchronized delivered-easing onsets

Outcome	Estimate	90% CI	Rotation ref.	Holm adj.	Rotations
Policy-ranked shadow carry spot return, months 0–1	-1.06	[-3.47, 1.17]	0.222	1.000	441
Carry-aligned CFTC non-commercial net share, change through month 3	-6.42	[-16.34, 4.17]	0.568	1.000	81
ACM 10-year expected-rate component, change through month 3	-0.13	[-0.31, 0.07]	0.202	1.000	396
ACM 10-year term premium, change through month 3	-0.03	[-0.19, 0.14]	0.980	1.000	396
OECD G7 CLI, change through month 6	-0.34	[-0.72, 0.00]	0.159	0.952	353
Chicago Fed NFCI, change through month 3	0.03	[-0.08, 0.15]	0.520	1.000	396
VIX, change through month 1	1.89	[-2.30, 5.81]	0.400	–	20

Notes: Events are synchronized delivered-easing onsets constructed from public BIS policy rates. They are a downstream policy-rate state, distinct from the paper’s predictive IYC state. Rotation-reference values enumerate every circular shift, condition on the observed number of valid episode outcomes, include the observed assignment, and double the smaller inclusive tail rank. Rotations reports the resulting finite reference count. The CFTC reference is small because missing contract coverage makes most rotations change the valid event count; its reference is correspondingly coarse. With irregular missingness, the same-count rotations need not form a transformation group, so these are conditional finite-rotation references rather than exact causal randomization tests. Intervals resample onset episodes. The Holm column adjusts the six declared primary reference values. Currency returns and CFTC positioning are percentage points, with CFTC positions measured as a share of open interest; ACM components are percentage points; CLI, NFCI, and VIX are index points.

The signs mostly fit a broad stress-and-easing chronology: the shadow carry proxy depreciates, carry-aligned positions decline, the expected-rate component falls, and the leading indicator weakens. Estimates are imprecise, and no primary outcome meets the 5-percent Holm-adjusted reference criterion. The U.S. term-premium estimate is nearly flat at three months. These results are directional mechanism evidence around policy delivery, not evidence that the inversion state predicts them.

Public proxies around synchronized delivered easing



Means across onset episodes; bands are 90% episode-bootstrap intervals. The delivered-easing state is downstream of, and distinct from, the predictive IYC state.

Figure 3: Outcomes around synchronized delivered-easing onsets. Event time zero is a month in which at least three policy rates fall by at least 10 basis points after a three-month quiet period. Panels show the code-defined cumulative return or endpoint-change estimands, with 90-percent episode-bootstrap intervals.

D.2.3 Sensitivity and interpretation

The shadow-carry response is recomputed using two- and four-country event thresholds and after omitting each event currency. Every threshold recomputes its own quiet period, so onset counts need not be monotone in the required number of cutting countries.

Table 14: Public-proxy sensitivity: shadow carry spot return

Construction	Estimate	Rotation ref.	Events	Rotations
At least three countries easing	-1.06	0.222	15	441
At least two countries easing	-1.10	0.120	22	434
At least four countries easing	-0.40	0.694	18	438
Exclude AUD from event rule	-0.56	0.515	17	439
Exclude CAD from event rule	0.35	0.548	18	438
Exclude CHF from event rule	-1.21	0.132	15	441
Exclude EUR from event rule	-1.02	0.182	16	440
Exclude GBP from event rule	-0.83	0.294	20	436
Exclude JPY from event rule	-0.24	0.838	19	437
Exclude NOK from event rule	-1.53	0.072	13	443
Exclude NZD from event rule	-0.53	0.590	15	441
Exclude SEK from event rule	-0.40	0.702	17	439

Notes: Each threshold definition recomputes its own three-month quiet window. Onset counts therefore need not be monotone in the number of countries required to cut. Leave-one-currency rows omit that currency from the event rule.

The exercise has four binding limits. Delivered easing is endogenous to prior information and occurs downstream of the paper’s predictor. Policy-rate ranks are not money-market or forward rates, and spot returns omit carry income, basis, transaction costs, and implementation lags. CFTC, ACM, and current-vintage macro series are imperfect proxies for the intended objects. Finally, fifteen onsets—fewer for some outcomes—leave low episode-level power. The public results are therefore useful as a reproducible challenge to mechanism implications, not as a replication of the headline state.

E Which countries matter in the public proxy?

This appendix asks whether particular country pairs or triples dominate the public delivered-easing association. The public policy-rate exercise does not reveal which curves created the baseline fifteen inversion episodes. The analysis uses the same BIS policy-rate and exchange-rate data as Appendix D.

E.1 Combination screen

All 36 currency pairs and 84 triples are enumerated. For a given combination, an onset occurs when every member cuts its policy rate by at least 10 basis points after three months without that same joint cut. The outcome is the public shadow-carry spot return accumulated over event months zero and one. The ranking of all nine currencies uses lagged policy-rate differentials; the tested country combination defines the event, not the portfolio.

Reference values are reported only for combinations with at least six valid events. Every eligible event sequence is shifted through every circular calendar rotation. Raw two-sided rotation references are supplemented with a common-rotation maximum absolute standardized reference across every eligible pair and triple. Under the maintained null that simultaneous cyclic shifts are exchangeable, the maximum reference controls the declared family; otherwise it is a finite family diagnostic. The 120-combination family was declared in code and exhaustively reported but was not preregistered. Episode-resampling intervals, leave-one-event means, and alternative minimum-event rules diagnose small-sample fragility.

Table 15: Country combinations in the public delivered-easing proxy

Combination	Events	Mean $[0, 1]$	Raw ref.	Max- $ z $ ref.
CHF-GBP	6	-4.54	0.009	0.035
AUD-EUR	6	-4.26	0.004	0.055
GBP-SEK	9	-3.32	0.013	0.119
EUR-NZD	6	-2.80	0.053	0.486
CHF-SEK	7	-2.74	0.070	0.508
CAD-NOK-NZD	7	-4.09	0.013	0.066
GBP-NZD-SEK	8	-3.71	0.018	0.086
NOK-NZD-SEK	6	-3.62	0.026	0.235
CAD-GBP-NOK	6	-3.33	0.031	0.321
AUD-NOK-NZD	6	-2.87	0.048	0.519

Notes: The table reports the five most negative eligible pairs and triples. An event requires every listed country to cut its BIS policy rate by at least 10 basis points after three months without the same joint cut. The outcome is the cumulative policy-ranked log spot-return proxy in event months 0 and 1, in percentage points. Portfolio legs target three currencies per side and expand to include all cutoff ties with equal weight. Combinations require at least six valid events. Raw reference values use every circular timing rotation. The final column is the common-rotation maximum absolute standardized reference across every eligible pair and triple. Under simultaneous cyclic-shift exchangeability it controls the declared family; otherwise it is a finite family diagnostic. The family was not preregistered. This is a delivered-easing proxy, not a replication of the yield-curve state.

Of 120 combinations, 42 have at least six events. CHF–GBP is the only combination meeting the 5-percent common-rotation criterion: six joint-cut onsets have a mean cumulative two-month shadow-carry log spot-return proxy of -4.54 percentage points, raw reference 0.009 , and maximum- $|z|$ reference 0.035 . The 90-percent episode-bootstrap interval is $[-9.52, 0.31]$, however, and deleting October 2008 reduces the mean to -2.52 . AUD–EUR is the next-most-negative pair, at -4.26 with maximum- $|z|$ reference 0.055 . CAD–NOK–NZD is the most-negative triple, at -4.09 with reference 0.066 .

The family result depends on admitting sparse combinations. With a six-event minimum, 42 combinations are eligible and one meets the 5-percent reference criterion. With minimums of eight or ten events, 25 and 15 combinations remain, no result meets that criterion, and the smallest maximum- $|z|$ reference values are 0.064 and 0.229 . The evidence does not establish a stable country bloc.

E.2 Sensor countries versus loss-bearing currencies

The public lead does not match the baseline loss-bearing cross-section. CHF and GBP have near-zero or modest equity betas and small, imprecise state-conditioned currency losses. AUD and NZD bear the larger losses. The own-curve horse race also shows that a country’s own curve generally does not forecast its currency after conditioning on the synchronized state.

This contrast suggests a testable distinction: countries whose bond or policy states reveal a shared episode need not be the currencies that bear its losses. Joint Swiss–U.K. easing may function as a calendar sensor for broad stress, while high-rate, high-beta currencies absorb the carry loss. The distinction is economically consistent with synchronization and exposure ordering, but the present result is only hypothesis-generating because delivered easing is downstream of forward-looking yield-curve inversions.

The direct test requires the nine-country live-state matrix. A fifteen-episode-by-nine-country contributor ledger could enumerate which curves activate each episode, then compare contributor combinations with the subsequent currency-loss cross-section under family-wide inference. The public combination screen specifies that future test but does not decompose the headline result.

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